Bayesian Inference, WEEK 01

Lecture Notes

1: Bayesian Inference , Classical Inference Overview

BAYESIAN INFERENCE 95% CI is $\left(\bar{X} - 1.96\frac{s}{\sqrt{n}},\ \bar{X} + 1.96\frac{s}{\sqrt{n}}\right)$ ✓

$(10, 20)$

Inference: Classical Inference

$$\begin{cases}\text{ESTIMATION} \begin{cases}\to \text{POINT ESTIMATOR} \\ \to \text{Confidence Interval}\end{cases}\end{cases}$$

$$X_n \overset{iid}{\sim} f(\cdot \mid \underline{\theta})$$

$$N(\underline{\mu}, 1) \quad \bar{X}_n \quad \boxed{\text{Var}(T(X))}$$

$$\text{ESTIMATOR} \to \text{MLE} \quad E(T(X)) \Big\} \to \text{MSE}$$

$$\boxed{X_1, \ldots, X_N} \quad \bar{X} = \frac{\sum X_i}{N} \quad \frac{\sum (X_i - \bar{X})^2}{N}$$

$$\frac{\sum (T^{(i)}(X_n) - \bar{T}_m)^2}{M} \quad \begin{pmatrix}\bar{Y}_1 \\ \vdots \\ \bar{Y}_M\end{pmatrix}$$

$$Y_1, \ldots, Y_M$$

$$\bar{X}_n \quad \text{Mean}(Y_1 \ldots Y_n) = \frac{\sum X_i}{n}$$

$$\begin{cases} Y_i & T^{(i)}(X_n) \\ \frac{\sum T^{(i)}(X_n)}{M} & \\ & T^{(1)}(X_n) \\ & \vdots \\ & T^{(M)}(X_n) = \bar{T}_M \end{cases}$$

2: Hypothesis Testing Issues, Practical vs Statistical Significance

$X_1, \ldots, X_n \overset{iid}{\sim} N(\theta, 1)$ $\quad \boxed{H_0 : \theta = 0 \text{ vs } H_1 : \theta \neq 0}$ $\quad H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$

$\alpha = 0.05$ $\quad H_0 : \boxed{p = 0.5}$

$\boxed{\text{Reject } H_0 \text{ if } \sqrt{n}\,|\bar{x}| > 1.96}$

Suppose true $\theta = \dfrac{10^{-10}}{10^{-5}}$ $\quad n = 10^{24}$ $\quad 10^{14}$. $\quad \sqrt{n}\,|\bar{x}| \approx 10^{12} \cdot 10^{-10} > 1.96$

0.50000 $\quad \boxed{\boxed{0.49, 0.51}}$ $\quad \pi\big(p \in (0.49, 0.51) \mid X\big)$

$\underline{\underline{\pi\big(\theta \in (-10^{-5}, 10^{5}) \mid X\big)}}$

$N(\mu, \sigma^2)$

Toss a coin, if head then $\boxed{n = 2}$, if tail then $\boxed{n = 1000}$ $\quad V(\bar{X}) = \dfrac{1}{2}\left[\dfrac{\sigma^2}{2} + \dfrac{\sigma^2}{1000}\right] \approx \boxed{\dfrac{\sigma^2}{4}}$

$\pi(\theta \mid \underline{\underline{\text{Data}}})$ $\quad (\theta - \frac{1}{2}, \theta + \frac{1}{2})$ $\quad \boxed{X_1 = 2,\ X_2 = 2}$

$U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ $\quad \boxed{X_2 - X_1}$ $\quad \checkmark$

3: Bayes Theorem , Disease Diagnostic Example

Bayes theorem: Suppose a disease affects 2% of a population.
A diagnostic test is available with the following properties:

- If a person has the disease, the test is positive with 0.95 prob.
- If a person doesn't have the disease, the test is positive with 0.10 prob. [false positive]

Let D = Event that a person has the disease.
T^+ : the test result is positive.
What is $\boxed{P(D \mid T^+)} = \frac{P(D \cap T^+)}{P(T^+)} = \frac{P(T^+|D)P(D)}{P(T^+)} = 0.1624$
Right column:

$$P(D) = 0.02 \quad P(D^c) = 0.98 \quad \text{[Prior probability]}$$
$$P(T^+ \mid D) = 0.95,\ P(T^+ \mid D^c) = 0.10 \quad \text{[Conditional probabilities]}$$
$$P(T^+) = P(D)P(T^+ \mid D) + P(D^c)P(T^+ \mid D^c) = 0.117 \quad \text{[Marginal prob.]}$$

4: Three-Prisoners Problem and Course Evaluation

Three-prisoners problem: $A \mid B \mid C$

$2 \rightarrow$ A asks the guard whether A will be executed? $P(c \mid b) = \frac{2}{3}$

$b \leftarrow B$ will be executed

$P(\text{Pardon for } A) = \frac{1}{3}$

$$P(\text{Pardon for } A \mid b) = (a)\frac{1}{3}\ (b)\frac{1}{2}\ (c)\frac{2}{3}\ (d)1 \quad P(\text{Pardon for } A \mid b) = \frac{P(A,b)}{P(b)}$$

$$\begin{aligned} P(A,b) &= P(A) \times P(b \mid A) \\ P(b) &= P(A)P(b \mid A) + P(B)P(b \mid B) + P(C)P(b \mid C) = \\ &= \frac{1}{3} \times \boxed{\frac{1}{2}} + \frac{1}{3} \times \boxed{0} + \frac{1}{3} \times \boxed{1} \\ P(A \mid b) &= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3} \times \frac{3}{2}} = \frac{1}{3} \end{aligned}$$

Course structure:

25% MID SEM.

50% END SEM $\rightarrow$ Assignment: Read and apply application paper

25% Project / Assignment / quiz Bayesian Data Analysis

(2) $\boxed{3-4}$ $(1-2)$ (Gelman)

(Stan) rstan, pystan

5: Prior, Likelihood, Posterior Revisited

$$\begin{aligned} P(D) &= 0.02 \text{ (prior)} \\ P(T^+ \mid D) &= 0.95 \text{ (conditional / likelihood)} \\ P(D \mid T^+) &= 0.1624 \text{ (Posterior)} \\ P(D) &= 0.20 \quad P(D^c) = 0.80 \quad P(D \mid T^+) = 0.704 \end{aligned}$$

$$P(D \mid T^+) = \frac{\boxed{P(D)}P(T^+ \mid D)}{\boxed{P(T^+)}} \qquad P(D^c \mid T^+) = \frac{P(D^c)P(T^+ \mid D^c)}{P(T^+)}$$

$\mathrm{P}(\mathrm{D} \mid T^+) \propto \underline{P(D)}\, P(T^+ \mid D)$ $\qquad P(D^c \mid T^+) = \dfrac{\boxed{P(D^c)}\boxed{P(T^+ \mid D^c)}}{k}$

$\mathrm{P}(\mathrm{D} \mid T^+) + P(D^c \mid T^+) = 1$ $\qquad P(D \mid T^+) = \frac{0.19}{k}$

$\boxed{P(D \mid T^+) \propto P(D)\, P(T^+ \mid D)}$ $\qquad \mathrm{P}(\mathrm{D}^c \mid T^+) = \frac{0.08}{k}$

$\boxed{\text{Posterior} \propto \underline{\text{prior}} \times \text{likelihood}}$

$$H_0 : p = 0.95 \text{ vs. } H_1 : p = 0.10 \quad P(T^+ \mid D) = 0.95 \quad P(T^+ \mid D^c) = 0.10$$

Prior (Subjective belief)

$$\boxed{\text{Prior}} \leftarrow \boxed{\begin{matrix} P(H_0) = 0.02 & P(H_1) = 0.98 \\ P(H_0) = 0.20 & P(H_1) = 0.80 \\ P(H_0) = 0.50 & P(H_1) = 0.50 \end{matrix}} \quad \boxed{P(H_0 \mid T^+) = 0.16 = 0.70}$$

$$\underline{\underline{\text{Cancer}}} \quad \underline{\underline{0.16}} \times \boxed{(C)}$$

6: Paradoxes in Classical Paradigm , Uniform Example

$$\underline{\underline{\text{Cancer}}} \quad \underline{\underline{0.16}} \times \boxed{(C)} \quad 0.16 \times C$$

$\underline{\text{Paradoxes in classical paradigm}}$ $\quad X \sim U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$

$$\boxed{\begin{matrix} [T_l(X), T_u(X)] \\ \theta \in [\bar{X} - \ldots, \bar{X} + \ldots] \rightarrow \boxed{\begin{matrix} 95\% \\ 50\% \end{matrix}} \\ \theta \in [1.5 - c, 1.5 + c] \text{ with } 75\% \text{ CI} \end{matrix}}$$

$$\begin{aligned} X_1, X_2 \; iid & \\ \text{let } y_1 &= \min\{X_1, X_2\} \\ y_2 &= \max\{X_1, X_2\} \\ \forall\theta, \; P(y_1 < \theta < y_2 \mid \theta) &= P(x_1 < \theta < x_2) + P(x_2 < \theta < x_1) \\ &= \tfrac{1}{2} \times \tfrac{1}{2} + \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{2}. \\ \therefore (y_1, y_2) & \text{ gives you a } 50\% \text{ CI of } \theta. \end{aligned}$$

Let $x_1 = 7.4, \; x_2 = 8.0 \quad (7.4, 8.0)$ $\quad 50\%$ CI of θ.

$\theta - \frac{1}{2} < x_1, x_2 < \theta + \frac{1}{2}$ $\quad$ We must have $\theta + \frac{1}{2} > 8.0 \Rightarrow \theta > 7.5$

$\therefore$ With certainty, $\theta \in (7.5, 7.9)$ $\quad \theta - \frac{1}{2} < 7.4 \Rightarrow \theta < 7.9$

7: Binomial vs Negative Binomial , Likelihood Principle

12 independent coin tosses are done and you get 9 heads.
$H_0 : p = \frac{1}{2}$ vs $H_1 : p > \frac{1}{2}$

Binomial: $L_1(p) = \binom{12}{9} p^9 (1-p)^3$

$\rightarrow$ p-value $= P(X \geq 9) = 0.075$

$\boxed{\alpha = 0.05}$

Negative Binomial:

$L_2(p) = \binom{11}{9} p^9 (1-p)^3$

$P(X \geq 9) = 0.0325$

Likelihood principle (LP): In making inferences or decision about θ, after x is observed, all relevant information is contained in the likelihood function of the observed x.

Advantages of Bayesian paradigm.

(1) Prior info can be incorporated

(2) Inferences conditional on actual data. $P(\theta)P(D_n \mid \theta)$

(3) LP

(4) Posterior $\rightarrow$ No need for separate hypothesis etc.

(5) Does not rely on asymptotic theory.

(6) However complex the model is, arbitrarily accurate inference by computational methods.

Bayesian Inference

Lectures 3–4: Steps of Bayesian Analysis, Exchangeability, and the Beta–Bernoulli Model

Lecture 3

1 The Three Steps of Bayesian Data Analysis

Overall goal

Bayesian data analysis is about *finding the conditional distribution of unknown (parameter) quantities*, given the observed data.

Step 1. Setting up a full probability model: construct a joint probability distribution for *all* observable and unobservable quantities, consistent with the knowledge about the underlying scientific problem and the data-collection process.

Step 2. Conditioning on observed data: the posterior distribution is the conditional distribution of the unobserved quantities, given the observed data.

Step 3. Evaluating the fit of the model.

To derive / write up

For Step 1, write out explicitly what "joint distribution for all observable and unobservable quantities" means in a simple running example (e.g. coin tosses with unknown bias θ): what is observable $(y_1, \ldots, y_n)$, what is unobservable (θ), and what is the joint $p(\theta, y_1, \ldots, y_n) = p(\theta)\, p(y_1, \ldots, y_n \mid \theta)$? For Step 3, note (from your course) what specific diagnostics/checks are used to evaluate model fit (e.g. posterior predictive checks) once they are formally introduced.

2 Posterior Odds vs. Prior Odds

Setup: choosing between two competing values/hypotheses for the parameter, θ_1 or θ_2 (e.g. testing whether an estimator is unbiased, or comparing two models).

Definitions

Prior odds: $\dfrac{p(\theta_1)}{p(\theta_2)}$, **Posterior odds:** $\dfrac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)}$.

Derivation (Bayes' rule applied to each hypothesis):

$$\frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)} = \frac{\dfrac{p(\theta_1)\, p(y \mid \theta_1)}{p(y)}}{\dfrac{p(\theta_2)\, p(y \mid \theta_2)}{p(y)}} = \frac{p(\theta_1)\, p(y \mid \theta_1)}{p(\theta_2)\, p(y \mid \theta_2)} = \underbrace{\frac{p(\theta_1)}{p(\theta_2)}}_{\text{prior odds}} \times \underbrace{\frac{p(y \mid \theta_1)}{p(y \mid \theta_2)}}_{\text{likelihood ratio (Bayes factor)}}.$$

Reading the comparison

$$\frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)} \gtrless \frac{p(\theta_1)}{p(\theta_2)} \iff \frac{p(y \mid \theta_1)}{p(y \mid \theta_2)} \gtrless 1,$$

i.e. the posterior odds move away from the prior odds in the direction favored by the likelihood ratio: if the data are more likely under θ_1 than θ_2, the posterior odds shift toward θ_1 relative to the prior odds, and vice versa.

As $n \to \infty$

The notes flag the behavior of $\dfrac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)}$ as the sample size n grows (written schematically as $n, n, n, n, \cdots \to$). With more and more data, the likelihood ratio should dominate the fixed prior odds, and the posterior odds should concentrate on whichever of θ_1, θ_2 is (closer to) the truth.

To derive / write up

State and prove this consistency result precisely: under repeated iid sampling, show that $p(y_1, \ldots, y_n \mid \theta_1)/p(y_1, \ldots, y_n \mid \theta_2) \to 0$ or ∞ a.s. depending on which hypothesis generated the data (a Kullback–Leibler / law-of-large-numbers argument on the log-likelihood ratio).

3 Conditional Independence and Exchangeability

Motivating example (from the notes): a sequence of observations $y_1, \ldots, y_{10}$ from 10 people, with an 11th person's observation y_{11} to be predicted (e.g. measurements in the range 100–200, such as heights in cm).

Conditional independence given θ

Given the parameter θ, a new observation is independent of the past observations:

$$p(y_{n+1} \mid \theta, y_1, \ldots, y_n) = p(y_{n+1} \mid \theta).$$

Marginally (i.e. before conditioning on θ), however,

$$p(y_{n+1} \mid y_1, \ldots, y_n) = f(y_1, \ldots, y_n)$$

is *not* simply $p(y_{n+1})$ — the past data inform beliefs about θ, which in turn inform the prediction of y_{n+1}. This is the basic mechanism of *learning from data* in the Bayesian framework.

Definition: Exchangeability

$y_1, \ldots, y_n$ are **exchangeable** if their joint distribution $p(y_1, \ldots, y_n)$ is *invariant with respect to permutation of the indices*: for any permutation π of $\{1, \ldots, n\}$,

$$p(y_1, \ldots, y_n) = p(y_{\pi(1)}, \ldots, y_{\pi(n)}).$$

In particular, for a new observation,

$$p(y_{n+1} \mid y_1, \ldots, y_n) = p(y_{n+1} \mid y_2, y_2, y_1, y_1, \ldots, y_n)$$

(the specific labeling/ordering of the conditioning variables does not matter).

Why this matters

Exchangeability is a weaker, more natural assumption than "iid" to justify from modeling considerations directly, yet (via de Finetti's representation theorem, Lecture 4) it turns out to be *equivalent* to the existence of a parameter θ and a prior $p(\theta)$ such that, conditional on θ, the y_i *are* iid. This is the conceptual bridge between "exchangeable data" and "iid-given-θ data with a prior on θ."

To derive / write up

Write out carefully why iid $\Rightarrow$ exchangeable but not conversely in general (an exchangeable sequence need not be iid *unconditionally* — only conditionally on θ under de Finetti's theorem). A short two-observation counterexample (e.g. draws without replacement from a finite population) is a good self-check.

Lecture 4

4 De Finetti's Representation Theorem

Setup: binary sequences, e.g. $1,0,0,1,0,\ldots$ or $1,1,0,0,0,\ldots$, with $\sum_{i=1}^{n} Y_i$ as the natural summary statistic.

De Finetti representation

If $y_1, y_2, \ldots$ is an (infinitely extendable) exchangeable sequence of binary outcomes, then there exists a distribution $p(\theta)$ such that

$$p(y_1,\ldots,y_n) = \int_0^1 \theta^{\sum_i y_i}(1-\theta)^{n-\sum_i y_i}\, p(\theta)\, d\theta.$$

Equivalently: $Y_1,\ldots,Y_n \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ for some random $\theta \sim p(\theta)$, and

$$\sum_{i=1}^{n} Y_i \,\Big|\, \theta \;\sim\; \text{Binomial}(n,\theta).$$

To derive / write up

This is the theorem that formally justifies "acting as if" the data are iid given a parameter θ, whenever all you are willing to assume is exchangeability. State the theorem's hypotheses precisely (infinite exchangeability, not just finite n) and note why finite exchangeability alone is *not* sufficient for the representation to hold exactly.

5 The Beta–Bernoulli Model

Model:

$$Y_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta), \qquad \mathbf{y} = (y_1,\ldots,y_n).$$

Prior – likelihood – posterior

$$\underbrace{p(\theta)}_{\text{prior}}, \qquad \underbrace{p(\mathbf{y}\mid\theta)}_{\text{likelihood}}, \qquad \underbrace{p(\theta\mid\mathbf{y})}_{\text{posterior}} \;\propto\; p(\theta)\,p(\mathbf{y}\mid\theta).$$

Beta density: for $X \sim \text{Beta}(a,b)$,

$$f_X(x) \;\propto\; x^{a-1}(1-x)^{b-1}, \qquad 0<x<1.$$

Posterior derivation

With likelihood $p(\mathbf{y}\mid\theta) = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$, and writing $z = \sum_i y_i$,

$$p(\theta\mid\mathbf{y}) \;\propto\; p(\theta)\,\theta^{z}(1-\theta)^{n-z}.$$

4.1 The Uniform Prior Special Case

Take $\theta \sim U(0,1)$, i.e. $p(\theta) = 1$ on $(0,1)$. Then

$$p(\theta\mid\mathbf{y}) \;\propto\; \theta^{z}(1-\theta)^{n-z} \quad\Longrightarrow\quad \theta\mid\mathbf{y} \;\sim\; \text{Beta}(z+1,\, n-z+1).$$

Marginal distribution of z under the uniform prior

With $p(\theta) = 1$, the marginal (prior predictive) distribution of $z = \sum_i Y_i$ is *discrete uniform* on $\{0, 1, \ldots, n\}$:

$$z \in \{0, 1, \ldots, n\}, \qquad \text{each with probability } \frac{1}{n+1}.$$

To derive / write up

Derive this marginal result directly:

$$p(z) = \binom{n}{z} \int_0^1 \theta^z (1-\theta)^{n-z}\, d\theta,$$

evaluate the integral using the Beta function $B(z+1, n-z+1) = \dfrac{z!\,(n-z)!}{(n+1)!}$, and simplify to confirm $p(z) = \dfrac{1}{n+1}$ for every $z \in \{0, \ldots, n\}$. This is a classical (and slightly surprising) special property of the uniform prior on a Bernoulli success probability.

4.2 Posterior Prediction

Goal: predict a new observation $\tilde{y}$ given the data $\mathbf{y}$.

Posterior predictive probability

$$\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \int_0^1 \Pr(\tilde{y} = 1 \mid \theta, \mathbf{y})\, p(\theta \mid \mathbf{y})\, d\theta = \int_0^1 \theta\, p(\theta \mid \mathbf{y})\, d\theta = \mathbb{E}[\theta \mid \mathbf{y}].$$

Using $\theta \mid \mathbf{y} \sim \text{Beta}(z+1, n-z+1)$ and the Beta mean formula $\mathbb{E}(X) = \dfrac{a}{a+b}$ for $X \sim \text{Beta}(a, b)$:

$$\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \mathbb{E}[\theta \mid \mathbf{y}] = \frac{z+1}{(z+1) + (n-z+1)} = \frac{z+1}{n+2}.$$

Laplace's rule of succession

$\dfrac{z+1}{n+2}$ is the classical "rule of succession": compare with the maximum likelihood estimate $\hat{\theta}_{\text{MLE}} = z/n$. The posterior mean shrinks the raw proportion slightly toward $1/2$, with the effect vanishing as $n \to \infty$.

To derive / write up

Verify $\dfrac{z}{n}$ vs. $\dfrac{z+1}{n+2}$ numerically for a few small (z, n) and confirm they converge as n grows. Also derive $\Pr(\tilde{y} = 1 \mid \mathbf{y}) = \mathbb{E}[\theta \mid \mathbf{y}]$ generally (not just for this conjugate case) by justifying the swap of integration order $\int p(\tilde{y} \mid \theta)\, p(\theta \mid \mathbf{y})\, d\theta$.

4.3 Summarizing the Posterior

For $\theta \mid \mathbf{y} \sim \text{Beta}(z+1, n-z+1)$, the standard summaries are:

Quantity	Notation
Posterior mean	$\mathbb{E}(\theta \mid \mathbf{y}) = \dfrac{z+1}{n+2}$
Posterior median	$\text{Med}(\theta \mid \mathbf{y})$
Posterior mode	$\text{Mode}(\theta \mid \mathbf{y})$
Posterior standard deviation	$\text{SD}(\theta \mid \mathbf{y})$
Maximum likelihood estimate	$\hat{\theta}_{\text{MLE}} = z/n$

The posterior density $\text{Beta}(z+1, n-z+1)$ is unimodal on $(0,1)$, rising from 0 at $\theta = 0$ to a peak and back down to 0 at $\theta = 1$ (sketch: a hump-shaped curve on $[0,1]$, skewed depending on z relative to n).

Credible interval percentiles noted in the margin

The values 0.05 and 0.95 are flagged — these are the standard lower/upper quantiles used to construct a (two-sided) 90% posterior credible interval for θ:

$$\left[\theta_{0.025},\ \theta_{0.975}\right] \text{ or } \left[\theta_{0.05},\ \theta_{0.95}\right],$$

depending on the convention (95% vs. 90%) being used — confirm which one your course defines.

To derive / write up

- Derive the mode of $\text{Beta}(a,b)$ for $a, b > 1$: $\text{Mode} = \dfrac{a-1}{a+b-2}$, and specialize to $a = z+1, b = n-z+1$.
- Derive the variance of $\text{Beta}(a,b)$: $\text{Var}(X) = \dfrac{ab}{(a+b)^2(a+b+1)}$, specialize similarly, and take the square root for $\text{SD}(\theta \mid \mathbf{y})$.
- Compare $\hat{\theta}_{\text{MLE}} = z/n$, the posterior mean $(z+1)/(n+2)$, and the posterior mode z/n (note: under a uniform prior the posterior *mode* coincides with the MLE, while the posterior *mean* does not) — confirm this coincidence algebraically.
- Numerically integrate (or use the incomplete Beta function) to compute the 0.05 and 0.95 quantiles of $\text{Beta}(z+1, n-z+1)$ for a specific (z,n) from class, to build intuition for how the credible interval narrows as n grows.

Lecture 5

The Beta–Bernoulli Conjugate Model

Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Ber}(\theta), \qquad \theta \sim U(0,1).$$

Then

$$p(\theta \mid x_1, \ldots, x_n) \sim \text{Beta}\left(\sum X_i + 1,\ n - \sum X_i + 1\right).$$

$$p(\theta \mid x_1 \ldots x_n) \ \propto\ \theta^{\sum X_i}(1-\theta)^{n-\sum X_i}$$

$$\text{Beta}(a,b) \ \propto\ \theta^{a-1}(1-\theta)^{b-1}$$

Conjugate prior. If the prior distribution is of the same family as the posterior distribution, then we call such a prior a **conjugate prior**.

$$p(\theta \mid X_1 \ldots X_n) \ \propto\ p(\theta)\, p(X_1 \ldots X_n \mid \theta)$$
$$p(\theta \mid X_1 \ldots X_n) \ \propto\ \theta^{\alpha+\sum X_i-1}(1-\theta)^{\beta+(n-\sum X_i)-1}$$
$$\therefore \quad \text{Beta}\Big(\sum X_i + \alpha,\ n - \sum X_i + \beta\Big)$$

$$p(\tilde{y} = 1 \mid X_1 \ldots X_n) = E(\theta \mid X_1 \ldots X_n) = \frac{\alpha + \sum X_i}{n + \alpha + \beta}$$

$$\text{As } n \to \infty, \quad \approx \frac{\sum X_i}{n}$$

General setup. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Ber}(\theta)$. If $p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$ [Prior], then $\theta \sim \text{Beta}(\alpha, \beta)$.

$$\boxed{\frac{\sum X_i}{n}} \qquad \text{Var}(\theta \mid X_1 \ldots X_n) = \frac{(\alpha + \sum X_i)(\beta + n - \sum X_i)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$$

$$\text{As } n \to \infty, \quad \text{Var}(\theta \mid X_1 \ldots X_n) \ \approx\ \frac{(\sum X_i)(n - \sum X_i)}{n^2(n)} \ \longrightarrow\ 0$$

Normal Distribution with Known Variance

$$p(y \mid \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$$p(\theta \mid y) \ \propto\ \underline{\underline{p(\theta)}} \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$$\exp\big(A\theta^2 + B\theta + C\big) \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$p(\theta) \sim N(\mu_0, \tau_0^2)$, & then,

$$p(\theta \mid y) \ \propto\ \exp\left(-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2\right) \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where}$$

$$\mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{1}{\sigma^2}y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \qquad \text{and} \qquad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

Show:

Now let $y_1, \ldots, y_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$.

$$\bar{y} \sim N\left(\theta, \frac{\sigma^2}{n}\right)$$

$$p(\theta \mid \bar{y}) \;\propto\; p(\theta)\, p(\bar{y} \mid \theta)$$

$$\therefore\ \theta \mid y_1, \ldots, y_n \text{ (equivalently } \theta \mid \bar{y})$$

$$\therefore\ \theta \mid \bar{y} \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \qquad \& \qquad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$p(y \mid \theta) \;\propto\; \exp\left\{-\frac{\sum(y_i - \theta)^2}{2\sigma^2}\right\} \qquad \exp\left\{-\frac{\left(\frac{\sum y_i}{n} - \theta\right)^2}{2\sigma^2/n}\right\}$$

$p(\theta)$ is $N(\underline{\mu_0}, \tau_0^2)$ $\qquad$ $p(\theta \mid y_1, \ldots, y_n)$ $\qquad$ $N(\mu_1, \tau_1^2)$

$\bar{y}$ is an estimator of θ.

Take $n = 1$:

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \qquad \boxed{\bar{y} = \bar{y}} \qquad \text{Precision} = \frac{1}{\text{Variance}}$$

- $\boxed{\mu_1 \equiv \mu_0}$ if $y = \mu_0$ or $\tau_0^2 = 0$ $\quad (\& \ \sigma^2 \neq 0)$
- $\mu_1 = y$ if $y = \mu_0$ or $\sigma^2 = 0,\ \tau^2 \neq 0$
- μ_1 is the weighted average of μ_0 & y.
- Let's take $\mu_0 = 0$, $\quad \mu_1 = \dfrac{y\,\tau_0^2}{\sigma^2 + \tau_0^2}$

$$\mu_1 = \frac{y\,\tau_0^2/n}{\sigma^2/n + \tau_0^2}$$

$$\tau_0^2(n) = \frac{\sigma^2}{n}$$

$$p(\theta \mid y) \;\propto\; p(\theta) \prod_{i=1}^{n} p(y_i \mid \theta)$$

Predictive Distribution

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\, p(\theta \mid y)\, d\theta$$

$$\propto \int \exp\left(-\frac{1}{2\sigma^2}(\tilde{y}-\theta)^2\right) \exp\left(-\frac{1}{2\tau_1^2}(\theta-\mu_1)^2\right) d\theta$$

pdf of bivariate normal $(\tilde{y}, \theta)$.

Q: $p(y) = ?$ (Marginal of y)

$$\theta \sim N(\mu_0, \tau_0^2), \qquad y \sim N(\theta, \sigma^2) \quad \text{then what is } p(y)?$$

Lecture 6

Recap and the Marginal Distribution of y

$$(\cdot \ \ldots X_n) \qquad \theta \mid x_1 \ldots x_n \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

$$p(y, \underline{\theta}) \ \propto \ \exp\left\{ -\begin{pmatrix}\theta \\ y\end{pmatrix}^T - \mu \right)^T \Sigma^{-1} \left(\begin{pmatrix}\theta \\ y\end{pmatrix} - \mu \right\}$$

$$\theta \sim N(\mu_0, \tau_0^2), \qquad y \mid \theta \sim N(\theta, \sigma^2)$$

$$\text{then} \quad p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta = \int \underbrace{\frac{\exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)}{\sigma\sqrt{2\pi}} \cdot \frac{\exp\left(-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2\right)}{\tau_0\sqrt{2\pi}}}\, d\theta$$

$$y \mid \theta \sim N(\theta, \sigma^2)$$

$$y \mid \theta = \theta + N(0, \sigma^2)$$

$$y = \theta + \varepsilon \qquad = \int p(y, \theta)\, d\theta$$

Marginal distribution of $\theta \sim N(\mu_0, \tau_0^2)$, y is Normal distribution

$$y \sim N(\mu_0,\ \tau_0^2 + \sigma^2)$$

$y = \theta + \varepsilon$ where $\theta \sim N(\mu_0, \tau_0^2)$.

$\begin{pmatrix}\theta \\ \varepsilon\end{pmatrix}$ is a bivariate normal r.v. $\qquad \varepsilon \sim N(0, \sigma^2)$ & θ, ε are independent.

The Joint (Multivariate Normal) Formulation

Let $y_1, \ldots, y_n \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ & $\theta \sim N(\mu_0, \tau_0^2)$. What is $p(y_1, \ldots, y_n)$?

$$y_1 = \theta + \varepsilon_1, \quad \ldots, \quad y_n = \theta + \varepsilon_n$$

$$\left.\begin{matrix} \varepsilon_1, \ldots, \varepsilon_n \text{ are iid } N(0, \sigma^2) \end{matrix}\right\} \text{independent of } \theta \sim N(\mu_0, \tau_0^2)$$

$$\begin{pmatrix}\theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n\end{pmatrix} \sim MVN(\mathcal{M},\ \Sigma), \qquad \mathcal{M} = \begin{pmatrix}\mu_0 \\ 0 \\ 0 \\ \vdots \\ 0\end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \tau_0^2 & 0 & & & \\ 0 & \sigma^2 & 0 & & \\ & 0 & \sigma^2 & 0 & \\ & & 0 & \sigma^2 & 0 \\ & & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix}y_1 \\ \vdots \\ y_n\end{pmatrix} = A \begin{pmatrix}\theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n\end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 0 & 0 & \cdots \\ 1 & 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 1 & \cdots \\ \vdots & & & & \ddots \end{pmatrix}, \qquad y = Ax$$

$$y = AX \implies y \sim MN\big(A\mathcal{M},\ A\Sigma A^T\big)$$

$$\mathrm{Cov}(\theta + \varepsilon_1,\ \theta + \varepsilon_2) = \tau^2$$

$$\theta \mid y_1, \ldots, y_n \sim N(\mu_1, \tau_1^2)$$

$$y \mid \theta \sim N(\theta, \sigma^2) \qquad \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim MN\left(\begin{pmatrix} \mu_0 \\ \vdots \\ \mu_0 \end{pmatrix}, \begin{pmatrix} \tau^2 + \sigma^2 & \tau^2 & \cdots \\ \tau^2 & \tau^2 + \sigma^2 & \\ \vdots & & \ddots \end{pmatrix} \right)$$

$$y \mid \theta = \underline{\underline{Z}} + \underline{\theta}, \qquad Z \sim N(0, \sigma^2)$$

Prediction for Unobserved Data

$$\theta \sim N(\mu_0, \tau_0^2)$$

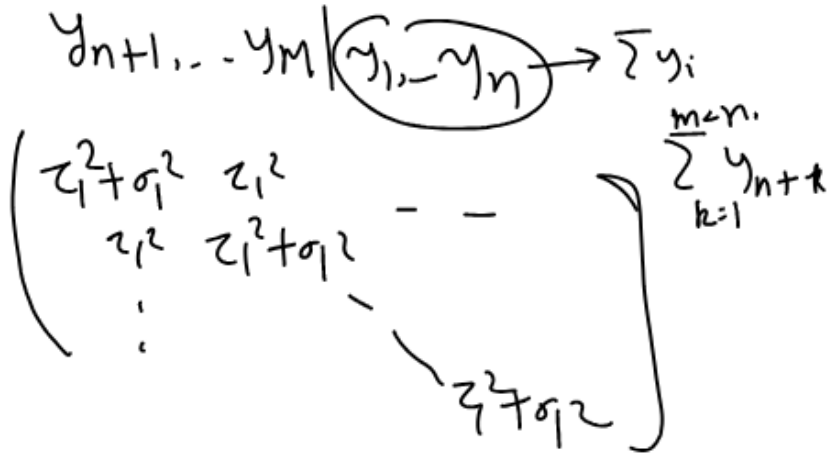
$$y_1, \ldots, y_M \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$$

Given $y_1, \ldots, y_n$, with y_M unobserved:

$$p(y_M \mid y_1, \ldots, y_n) = \int \underline{p(y_M \mid \theta)}\ \underline{\underline{p(\theta \mid y_1, \ldots, y_n)}}\ d\theta \qquad y_M \mid y_1, \ldots, y_n \sim N\big(\mu_1,\ \sigma^2 + \tau_1^2\big)$$

$$\int p(y \mid \theta)\, \underline{\underline{p(\theta)}}\, d\theta \qquad y_{n+1}, \ldots, y_M \mid \underbrace{y_1, \ldots, y_n}\ \longrightarrow \sum_{k=1}^{M-n} y_{n+k}$$

$$\tau_1^2 = \frac{1}{\dfrac{1}{\tau_0^2} + \dfrac{n}{\sigma^2}} \longrightarrow 0 \qquad \Big(\text{so } p(y_1, \ldots, y_n) \longrightarrow \delta_{\theta_0} \text{ as } n \to \infty\Big)$$



The Beta–Bernoulli Model Revisited (General α, β)

$$X_1, \ldots, X_n \mid \theta \overset{\text{iid}}{\sim} \text{Ber}(\theta), \qquad \theta \sim \text{Beta}(\alpha, \beta)$$

$$P(X_1 = 1) = \frac{\alpha}{\alpha + \beta}$$

$$P(X_1, \ldots, X_n) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{\sum X_i + \alpha - 1} (1-\theta)^{n - \sum X_i + \beta - 1} \, d\theta = \frac{B(\alpha + \sum X_i,\ \beta + n - \sum X_i)}{B(\alpha, \beta)}$$

A Note on Independence

$$P(X_1 = 1,\, X_2 = 1) \ \neq \ P(X_1 = 1)\, P(X_2 = 1)$$

$$\begin{matrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{matrix}$$

Sampling algorithms (mentioned in passing):

- Metropolis–Hastings algorithm
- MCMC (Markov Chain Monte Carlo)
- HMC (Hamiltonian Monte Carlo)

Exponential Family of Distributions

$p(y_i \mid \theta) = f(y_i)\, g(\theta)\, \exp\left\{\phi(\theta)^T u(y_i)\right\}$. e.g. Poisson, Normal, Binomial, Exponential.

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} f(y_i)\, (g(\theta))^n \, \exp\left\{\phi(\theta)^T \sum_{i=1}^{n} u(y_i)\right\}.$$

$$p(\theta \mid y) \propto p(\theta)[(g(\theta))^n \exp\left\{\phi(\theta)^T \sum_i u(y_i)\right\}].$$

$$p(\theta) \propto (g(\theta))^\eta \exp\left\{\phi(\theta)^T \nu\right\}.$$

$$\text{Then,} \qquad p(\theta \mid y) \propto (g(\theta))^{\eta+n} \exp\left\{\phi(\theta)^T \left(\sum_i u(y_i) + \nu\right)\right\}.$$

Show that the Binomial is an exponential family of distribution. Hence put an appropriate conjugate prior over it.

$$\text{Beta}(\alpha, \beta) \qquad \boxed{\frac{\alpha}{\alpha + \beta}}$$

$$\left[\text{Natural parameter is } \log(\theta/(1-\theta))\right]$$

BI-W04

Lec 7, August 11

Conjugate prior: Normal–Normal (mean unknown, variance known)

$\theta \sim N(\mu_0, \tau_0^2)$. Should have high variance.

$$100\,\text{cm} - 200\,\text{cm}, \qquad Y_1, \dots, Y_n \mid \theta \sim N(\theta, \sigma^2)$$

$$\pi(\theta \mid Y) \propto \frac{1}{\sqrt{2\pi\tau_0^2}} \exp\Big\{-\frac{(\theta-\mu_0)^2}{2\tau_0^2}\Big\} \cdot \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\Big\{-\frac{\sum(Y_i-\theta)^2}{2\sigma^2}\Big\}$$

As $\tau_0 \to \infty$,

$$\pi(\theta \mid Y_1, \dots, Y_n) \approx \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\Big\{-\frac{(\theta-\bar{Y})^2}{2\sigma^2/n}\Big\}$$

$$p(Y_1, \dots, Y_n) = p(Y_1, \dots, Y_n \mid \theta)\,\pi(\theta)$$

Examples of priors:

$$N(0, 100), \qquad N(0, 10), \qquad (Y_1, \dots, Y_n)$$

Regression illustrations:

$$y = \beta_0 + \beta_1 x + \varepsilon$$
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

Does it need to be a valid probability density function?

Answer

Prior: **No**

Posterior: **Yes**

$$\pi(\theta \mid y) \propto \pi(\theta)\,\pi(y \mid \theta) \qquad \pi(\theta \mid x) = \frac{\pi(\theta)\,\pi(x \mid \theta)}{\int \pi(\theta)\,\pi(x \mid \theta)\,d\theta}$$

$\pi(\theta \mid A) \propto \pi(\theta)\,\pi(A \mid \theta)$ need not integrate to 1.

But

$$\int \pi(\theta \mid x)\,d\theta = 1 \quad \text{if} \quad \int \pi(\theta)\,\pi(x \mid \theta)\,d\theta \text{ is finite } (< \infty).$$

Improper prior

$$\frac{\pi(\theta)}{\int_{-\infty}^{\infty} \pi(\theta)\,d\theta} \propto 1 \quad \Rightarrow \quad \int \pi(\theta)\,\pi(x \mid \theta)\,d\theta = \infty$$

$$\int \pi(\theta)\,\pi(x \mid \theta)\,d\theta < \infty$$

$$\int \pi(\theta)\, d\theta = C \leq \infty \quad \Rightarrow \quad \int \pi(\theta)\, \pi(x \mid \theta)\, d\theta \leq \infty$$

$$\frac{1}{C} \int \pi(\theta)\, \pi(x \mid \theta)\, d\theta \qquad \int \pi(\theta, x) d\theta = 1$$

Example 1

$$X \mid \theta \sim N(\theta, 1), \qquad \pi(\theta) \propto 1 \quad \Rightarrow \quad \int \pi(\theta)\, d\theta = \infty.$$

But after observing X,

$$\pi(\theta \mid x) \propto \pi(\theta)\, f(x \mid \theta) \quad \Rightarrow \quad \pi(\theta \mid x) \propto \exp\Big\{-\frac{(\theta - x)^2}{2}\Big\}.$$

$$\theta \mid x \sim N(x, 1)$$

Example 2

$$X \mid \theta \sim U(0, \theta), \qquad \pi(\theta) \propto 1.$$

$$\theta \sim N(\mu_0, \tau_0^2) \quad (\theta = \mu_0), \qquad \frac{1}{\sqrt{2\pi\tau_0^2}} \exp\Big\{-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\Big\}$$

$$\pi(\theta) \propto 1$$

$$\pi(x \mid \theta) = \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

$$\pi(\theta \mid x) \propto \pi(\theta)\pi(x \mid \theta) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

$$\pi(\theta) \propto \frac{1}{\theta} \quad \longrightarrow \quad \text{proper posterior,}$$

$$\pi(\theta) \propto 1 \quad \longrightarrow \quad \text{improper posterior.}$$

$$\int \pi(\theta)\pi(x \mid \theta)\, d\theta < \infty.$$

$$\theta \sim N(\mu_0, \sigma_0^2).$$

$$\theta = \mu_0, \theta = \mu_0 + 10$$

$$\pi(\mu_0) = \frac{1}{\sigma_0\sqrt{2\pi}}.$$

$$\pi(\mu_0 + 10) = \frac{1}{\sigma_0\sqrt{2\pi}} \exp\left\{-\frac{10^2}{2\sigma_0^2}\right\}.$$

$$\frac{\pi(\mu_0)}{\pi(\mu_0 + 10)} = \exp\left(\frac{10^2}{2\sigma_0^2}\right).$$

$$N(0, 10), \qquad N(0, 100), \qquad p(y_1, \ldots, y_n)$$

$$\int \pi(\theta)\, d\theta \neq 1$$

$$\theta \sim N(0, 10) \quad \text{or} \quad \theta \sim N(0, 100)$$

Further

$$Y_1, \ldots, Y_n \mid \sigma^2 \sim N(\mu, \sigma^2), \quad \mu \text{ known.}$$

What is $p(\sigma^2)$? (conjugate prior)

Binomial (Derive conjugate prior) → Express in terms of exponential family.
Poisson (Derive conjugate prior) → Express in terms of exponential family.

Derive $P(\lambda)$ using the representation of Poisson in exponential family of distributions.

$$\pi(\theta) \propto \exp\Big\{\eta(\theta)^T T(x) - A(\theta)\Big\}$$

Hints

- Identify natural parameter, sufficient statistic.
- Construct prior over these natural parameters.
- Use Jacobian to convert that prior for the usual parameter of the distribution.
- (Hint: Think along the meaning of Beta prior for Bernoulli.)

Lecture 8, August 12, 2026

Poisson Regression with Covariate

Poisson regression with covariate.

$$(x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_n, y_n)$$

$$(y_1, y_2) \qquad \text{Priors \& Likelihood}$$

$$y_i \mid x_i \sim \text{Poisson}(\theta x_i)$$

$$p(y_1, \dots, y_n \mid x_1, \dots, x_n, \theta)$$

$$L(\theta) = \prod_i \frac{e^{-\theta x_i}(\theta x_i)^{y_i}}{y_i!}$$

$$L(\theta) \propto e^{-\theta \sum x_i}\, \theta^{\sum y_i}$$

$$L(\theta) \propto e^{-\theta \sum x_i}\, \theta^{\sum y_i}$$

1) Likelihood
2) Conjugate prior (or some other form)
3) Posterior
4) Predictive distribution

Non-informative Prior

$$\pi(\theta) \propto 1 \qquad X \mid \theta \sim N(\theta, \sigma^2) \qquad \pi(\theta) \propto 1 \qquad \pi(y \mid \theta) \propto 1$$

$$\frac{1}{\sigma.\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$\frac{1}{\sigma.\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{where } \theta = e^{\gamma}$$

Non-informative prior:

$$\phi = g(\theta) \qquad \pi(\phi) = \pi(g^{-1}(\phi))\,|J|$$

Jeffreys Prior

Jeffreys prior:
Any rule for determining the prior density should yield an equivalent result if applied to the transformed parameter, i.e. $p(\phi)$ computed by determining θ and applying (1) should match with the distribution that is obtained directly for $p(\phi)$.
Let $\phi = h(\theta)$ where h is a one-to-one transformation. Then,

$$p(\phi) = p(\theta) \left| \frac{d\theta}{d\phi} \right| = p(\theta) \left| h'(\theta) \right|^{-1}$$

Jeffreys principle says that $\pi(\theta) \propto \sqrt{I(\theta)}$ where $I(\theta)$ is the Fisher information for θ, i.e.

$$I(\theta) = -E\left(\frac{\partial^2 \log f(y \mid \theta)}{\partial \theta^2} \right)$$

$$\pi(\theta) \propto \sqrt{I(\theta)} = \sqrt{I(\phi)} \left| \frac{d\phi}{d\theta} \right|$$

Then,

$$I(\phi) = -E\left(\frac{\partial^2 \log f(y \mid \phi)}{\partial \phi^2} \right)$$

$$E\left[\left(\frac{\partial \log f(y \mid \theta)}{\partial \theta} \right)^2 \right] = I(\theta) \left| \frac{d\theta}{d\phi} \right|^2$$

$$\theta \to \theta + d\theta$$

$$\theta \in [\theta, \theta + d\theta] \;\to\; p(\theta) \approx 0$$

$$X \to p \qquad p > 0.5 \qquad \mathrm{Var}(X) = \sqrt{\frac{N(1-p)}{p}}$$

$$p(\theta \mid X) \propto p(X \mid \theta + d\theta)$$

$$p(X \mid \phi) \propto p(X \mid \phi + d\phi)$$

Binomial

$$p = 0.5 \qquad X \mid \theta \sim \mathrm{Bin}(n, p) \qquad \theta = \frac{p}{1-p} \qquad \pi(\theta) \propto 1$$

Binomial

$$-E\left(\frac{\partial^2 \log f(y \mid \theta)}{\partial \theta^2} \right) = \frac{1}{p(1-p)}$$

Kullback–Leibler Divergence

K–L divergence (Kullback–Leibler divergence)

$$D_{KL}(P \parallel Q) = E_P\left[\log \frac{p}{q}\right]$$

$$E_\theta[d \log f(y \mid \theta) - log f(X|\theta) d\theta]$$

$$\theta \to \theta + d\theta$$

$$d\theta^2 = I(\theta)\, d\theta^2$$

$$L(\theta) = \log f(y \mid \theta)$$

$$L(\theta + d\theta) = \log f(y \mid \theta + d\theta)$$

$$L(\theta + d\theta) = L(\theta) + L'(\theta)\, d\theta + \tfrac{1}{2} L''(\theta)\, d\theta^2 + o(d\theta^2)$$

$$-E_\theta[L'(\theta)]\, d\theta$$

$$-\tfrac{1}{2} E_\theta[L''(\theta)]\, d\theta^2 + o(d\theta^2)$$

$$D_{KL}\big(P(X|\theta), P(X|\theta + d\theta)\big) = 0 + I(\theta)(d\theta)^2 + o(d\theta^2)$$

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

Poisson Example

$$\frac{e^{-\lambda} \lambda^x}{x!}$$

$$I(\lambda) = \frac{1}{\lambda} \qquad \sqrt{I(\lambda)} = \frac{1}{\sqrt{\lambda}}$$

$$0 \to 2.25 \qquad 2.25 \to 6.25$$

$$\sqrt{\lambda}$$

Lec 9

Tuesday, August 18, 2026 2:32 PM

40 mins quiz on 24th August

4 – 4:40pm.

5% of the course.

(−25/100) penalty for cheating. (If your final marks at the end of the course is 50/100. then ——→ 25/100).

$\pi_J(\theta) \propto \sqrt{I(\theta)}$

$\phi = h(\theta)$

$\pi_J(\phi) \propto \sqrt{I(\phi)}$

$E\left(\frac{\partial^2 \log L}{\partial \theta^2}\right)$

$\phi = h(\theta)$

$I(\phi) = I(\theta)\left(\frac{d\theta}{d\phi}\right)^2$

$p(\theta)$

CLRB(b)

b

Bin(n, p)

p

$\text{Var}(T(X)) \geq \frac{1}{I(\theta)}$

$\theta \leftarrow T(X)$ $\quad T(X) = \bar{X}$

$I(\theta)$

$\text{Var}(T(X)) \geq \frac{p(1-p)}{n}$

Poisson(λ)

$\lambda \in \mathbb{R}^+$

① Derive Jeffreys prior for Poisson(λ).

1 2 100 101

$\pi_J(\lambda) \propto \frac{1}{\sqrt{\lambda}}$

$\int_1^2 \pi_J(\lambda)\, d\lambda$

$KL(p_\lambda, p_{\lambda+d\lambda}) = \frac{1}{8}(\pi_J(\lambda))^2 (d\lambda)^2$

② Derive for Binomial.

(b) $\psi = \log \frac{p}{1-p}$

0 1

What is k: $\int_1^k \pi_J(\lambda)\, d\lambda = \int_0^1 \pi_J(\lambda)\, d\lambda$

$\int \pi_J(\lambda)\, d\lambda = 2\sqrt{\lambda}$

$\int_0^p \pi_J(p)\, dp$.

$\sin^{-1}(\sqrt{p})$

$N(\mu, \sigma^2)$ σ^2 known.

$\pi_J(\mu) \propto 1$

$X = Z + \mu$ where $Z \sim N(0, \sigma^2)$

Bin: $\pi_J(p) \simeq \text{Beta}(\frac{1}{2}, \frac{1}{2})$.

Poisson: $\pi_J(\lambda) = \text{Gamma}(\frac{1}{2}, 0)$

$X = Z + \mu$, $N(0, \sigma^2)$

Normal: known σ^2 : $\pi_J(\mu) \propto 1$. unknown μ

$\pi_J(\nu)$

$\nu = \mu^3$ $\quad N(\nu^{1/3}, \sigma^2)$

$\pi(\mu) \propto 1$

100 cm – 250 cm

Non-informative prior: Jeffreys prior.

Flat prior: $\pi(\theta) \propto 1$.

Diffuse/Vague prior: $N(\mu_0, \tau_0^2)$ where $\tau_0^2 \to \infty$.

Normal mean known. variance unknown $\pi_J(\sigma^2) \propto$.

Multi-parameter models

0 1 Bin(n, p)

Wednesday, August 19, 2026 2:03 PM

Multi-parameter models.

(θ_1, θ_2) $\pi(\theta_1, \theta_2)$ ↓ Prior.

$X_1, \ldots, X_n \overset{iid}{\sim}_{\theta_1, \theta_2} f(x; \theta_1, \theta_2)$

e.g. $\theta_1 = \mu$, $\theta_2 = \sigma^2$.

$X_1, \ldots, X_n \overset{iid}{\sim}_{\mu, \sigma^2} N(\mu, \sigma^2)$

$\pi(\mu, \sigma^2)$ ✓

$$\pi(\theta_1, \theta_2 | D) \propto \pi(\theta_1, \theta_2)\, \pi(D | \theta_1, \theta_2)$$

μ, σ^2 → Nuisance parameters.

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D)\, d\theta_2$$

$\pi(\theta_1 | D)$ Monte-Carlo integration

(X, Y) $X^{(1)}, Y^{(1)}$ $X^{(1)}, Y^{(2)}$ $X^{(B)}, Y^{(B)}$

$\pi(X, Y) = \pi(X|Y)\pi(Y)$

$\theta_1^{(1)}, \theta_2^{(1)}$
⋮
$\theta_1^{(B)}, \theta_2^{(B)}$

Marginal of $\theta_1 | D$ can be obtained by ignoring the samples of θ_2

$$\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D)\, d\theta_2 = \int \pi(\theta_1 | \theta_2, D)\, p(\theta_2 | D)\, d\theta_2$$

$\pi(\theta_1 | \theta_2, D)$ $\pi(\theta_2 | D)$

Single-parameter inference.

$\theta_2 \to M_1, M_2$ → $\pi(\theta_1 | M_1, D)$

$\pi(\theta_2 | D)$

Suppose you want to estimate μ & σ^2 in $N(\mu, \sigma^2)$ model.
For known value of σ^2 you know how to estimate.
and you also know. $p(\sigma^2 | D)$ $\pi(\mu, \sigma^2 | D)$

$$\pi(\mu | D) = \int \pi(\mu | \sigma^2, D)\, \pi(\sigma^2 | D)\, d\sigma^2$$

σ^2 (known) — this needs to be calculated

Generating sample from $\pi(\mu | D)$ can be done in 2-steps:

Independent

Step1: Generate samples from $\sigma^2 | D$ (call it $\sigma^{2(s)}$).

Step2: $\mu | \sigma^{2(s)}, D$.

Gibbs sampling: $\sigma^2 | \mu, D \to \sigma^{(s)}$ → conditional.
$\mu | \sigma^2, D \to \mu^{(s)}$ → conditional
→ Dependence.
$\sigma^{2(s+1)} | \mu^{(s)}, D$
$\mu^{(s+1)} | \sigma^{2(s+1)}, D)$

in Gibbs sampler.

Go to step B & Iterate

(X, Y) $\pi(X)$ → Generate from marginal dist^n of X

& then $\pi(Y|X)$ → conditional dist^n

Repeat

Normal unknown σ^2, known μ.

Scaled-inverted-χ^2 distribution:

$$f(x; \nu, \tau^2) = \tau^{\nu} \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \frac{e^{-\frac{\nu\tau^2}{2x}}}{x^{(\frac{\nu}{2}+1)}}, \quad x > 0$$

Scaled-inv-$\chi^2(\nu, \tau^2)$. ν = degrees of freedom. τ^2 is scale

If $X \sim$ Scaled-inv-$\chi^2(\nu, \tau^2)$.

then a) $\frac{\nu\tau^2}{X} \sim \chi^2(\nu)$ b) $X \sim$ inv-Gamma$\left(\frac{\nu}{2}, \frac{\nu\tau^2}{2}\right)$

c) $kX \sim$ Scaled-inv-$\chi^2(\nu, k\tau^2)$

$\left[\frac{\nu\tau^2 k}{kX} \sim \chi^2(\nu) \Rightarrow kX \sim \text{Scaled-inv-}\chi^2(\nu, k\tau^2)\right]$

Let $X_1, \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ μ is known.

Let $\sigma^2 \sim$ Scaled-inverse-$\chi^2(\nu_0, \sigma_0^2)$.

Then, $\sigma^2 | D \sim$ Scaled-inverse-$\chi^2(\nu, s^2)$

where $\nu = \nu_0 + n$, $s^2 = \nu_0 \sigma_0^2 + \Sigma(x_i - \mu)^2$.

$\nu_0 + n$

$\sigma^2 \sim$ Scaled-inverse-$\chi^2(\nu_0, \sigma_0^2)$.

$Y|\sigma^2 \sim N(\mu, \sigma^2)$ μ is known

Then, find the marginal dist^n of Y.

$Y \sim t_{\nu_0}(\mu, \sigma_0^2)$ → DO IT

(2)

$p(y) = \int p(y|\sigma^2)\pi(\sigma^2)\,d\sigma^2$

$y|\sigma^2 \sim N(\mu, \sigma^2)$

σ^2 → known.

μ → unknown & μ follows normal

↑ Marginal of Y is normal.

$\sigma^2|D \sim$ Scaled-inv-$\chi^2(\nu_n, s_n^2)$

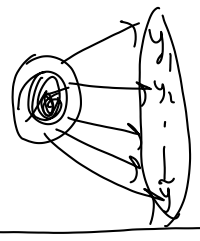
(3) $p(\tilde{y}|D)$ (Posterior predictive distribution).

$\tilde{y}_1 \dots \tilde{y}_n \overset{iid}{\sim} N(\mu, \sigma^2)$ μ known.

$\sigma^2 \sim$ Scaled-inv-$\chi^2(\nu_0, \sigma_0^2)$

$\sigma^2|D \sim$ Scaled-inv-$\chi^2(\nu_n, s_n^2)$

$\tilde{y}|\sigma, D)$



$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$ → Find.

→ Scaled-inverse-χ^2

y

$y_1, \dots y_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$p(\sigma^2|y)$?

Course Notes: Lecture 9 & Multi-Parameter Models

CLAD IV

August 2026

Lecture 9: Overview & Administration

Lecture Details & Quiz Logistics

- **Date/Time:** Tuesday, August 10, 2026, 2:32 PM
- **Course:** CLAD IV
- **Quiz Schedule:** 40 mins quiz on 24th August, 4:00 PM – 4:40 PM
- **Weightage:** 5% of the course
- **Policy Note:** $(-25/100)$ for cheating (if you're fined, marks at the end of the course are safe). Then 25/100.

Fisher Information & Prior Derivations

- Loss / Function relations: $G(\cdot), \quad L\left(\frac{\theta}{\hat{\theta}}\right)$
- $\Phi_i = -f_i(\theta)$
- Prior proportionality: $\pi_T(\theta) \propto \sqrt{I(\theta)}$
- Information identity: $E\left(\frac{\partial^2 \log L}{\partial \theta^2}\right) = -I(\theta)$
- Reparameterization ($\phi = \text{fn}(\theta)$):

$$I(\phi) = I(\theta)\left(\frac{d\theta}{d\phi}\right)^2$$

- Variance bound: $\text{Var}(T(x)) \geq \frac{1}{I(\theta)}$
- Variance and distributions: $\text{Var}(\cdot)$, $\text{Poisson}(\cdot)$, $\frac{k(1-k)}{n}$

Jeffreys' Prior Derivations

Poisson Distribution:

$$\text{Derive Jeffreys' prior: } \pi_J(\lambda) \propto \frac{1}{\sqrt{\lambda}}$$

$$\int_k^{\mathbb{R}} \pi_T(x)|dx|, \quad k_L(\theta_\lambda, k_\lambda d\lambda) = k_L(\pi\, ds)^2 dt$$

Binomial Distribution:

$$\text{Derive for Binomial: } \theta = \frac{p}{1-p} \implies \phi = \log \frac{p}{1-p}$$

Miscellaneous Notes & Vector Representations

- Matrix / Column Vector structure:

$$\begin{pmatrix} A_x \\ X_y \\ X_y \\ X_y \\ X_y \\ X_y \\ X_z \\ Y_z \end{pmatrix}$$

- Differential relation: $(x, \Omega)d\lambda = 2\sqrt{3}$
- $F \to (x, x^2)$ Brown $\int \frac{P}{\pi_T}(p) d^2P$
- $\pi(\theta) = \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$
- Transformation: $x = z + (1 - x)$ where $z \sim N(0, s^2)$
- $\pi_1(\nu) = \frac{e}{\text{Jeffreys}}\left(\frac{1}{2}, \nu\right) \quad \text{NIU} \quad x = z + \boxed{\rho 1}$
- $\pi_\xi\langle\cdot\rangle \quad \left(-1(\sqrt{n^2}, e^2)\right)$

Priors Taxonomy

- **Non-informative / Jeffreys prior**
- **Flat prior:** $\pi(\theta) \propto 1$
- **Diffuse prior:** $(\pi_0(\cdot))$
- **Normal mean known, unknown σ^2:** $\pi(\sigma^2) \propto \frac{1}{\sigma^2}$
- **Multi-parameter models:** $2\sin^{-1}(\sqrt{p}), \quad [b_1, m_1, p_2]$

Multi-Parameter Models

Wednesday, August 19, 2026 – 2:05 PM

1. Setup and Nuisance Parameters

- Prior distribution over parameters: $\pi(\theta_1, \theta_2)$
- Data distribution: $X_1, \ldots, X_n \mid \theta_1, \theta_2 \sim f(x; \theta_1, \theta_2)$
- Example (Normal Model): $X_1, \ldots, X_n \mid \mu, \sigma^2 \sim N(\mu, \sigma^2)$
- Joint posterior distribution:

$$\pi(\theta_1, \theta_2 \mid D) \propto \pi(\theta_1, \theta_2)\, \pi(D \mid \theta_1, \theta_2)$$

- **Nuisance Parameters (Marginalization):**

$$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D)\, d\theta_2$$

$$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D)\, d\theta_2 = \int \pi(\theta_1 \mid \theta_2, D)\, \pi(\theta_2 \mid D)\, d\theta_2$$

Monte Carlo Integration & Sampling

Sampling sequence: $(X^{(1)}, Y^{(1)}), (X^{(2)}, Y^{(2)}), \ldots$ where $\pi(x, y) = \pi(x \mid y)\pi(y)$.

- Samples generated: $\theta_1^{(g)}, \theta_2^{(g)}$
- The marginal distribution of θ_1 can be obtained simply by ignoring the samples of θ_2.
- $\pi(\theta_2 \mid D)$: $\quad \theta_2 \to (M_1), M_2 \implies \pi(\theta_1 \mid M_1, D)$ (Single-parameter inference).

2. Estimation in $N(\mu, \sigma^2)$ Model

Suppose you want to estimate μ & σ^2 in $N(\mu, \sigma^2)$ model. For a known value of σ^2, you know how to estimate, and you also know $p(\sigma^2 \mid D)$.

$$\pi(\mu \mid D) = \int \pi(\mu \mid \sigma^2, D)\, \pi(\sigma^2 \mid D)\, d\sigma^2 \quad \text{(to be calculated)}$$

Sampling Algorithms

Generating samples from $\pi(\mu \mid D)$ in 2 steps:

Step 1: Generate samples from $\pi(\sigma^2 \mid D)$ (call it $\sigma^{2(g)}$).

Step 2: Generate sample from conditional distribution $\pi(\mu \mid \sigma^{2(g)}, D) \to \mu^{(g)}$.

Gibbs Sampling (Conditional):

- $\sigma^2 \mid \mu, D \to \sigma^{2(g)}$
- $\mu^{(g+1)} \mid \sigma^{2(g+1)}, D \to \mu^{(g+1)}$
- Go to Step 1 & Iterate.

3. Scaled-Inverse-χ^2 Distribution

Density function:

$$f(x;\nu,s^2)=\frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)}\frac{(s^2)^{\nu/2}}{x^{(\nu/2+1)}}e^{-\frac{\nu s^2}{2x}},\quad x>0$$

where ν represents degrees of freedom, and s^2 is the scale parameter.

Properties of Scaled-Inv-$\chi^2(\nu$

If $X\sim$ Scaled-inv-$\chi^2(\nu,s^2)$, then:

a) $\frac{\nu s^2}{X}\sim\chi^2(\nu)$

b) $X\sim$ Inv-Gamma $\left(\frac{\nu}{2},\frac{\nu s^2}{2}\right)$

c) $kX\sim$ Scaled-inv-$\chi^2(\nu,ks^2)$

Proof of (c):

$$\frac{\nu s^2 k}{kX}\sim\chi^2(\nu)\implies kX\sim\text{Scaled-inv-}\chi^2(\nu,ks^2)$$

4. Conjugate Inference & Predictive Distributions

Normal model with unknown σ^2, known μ:

- Let $X_1,\ldots,X_n\sim N(\mu,\sigma^2)$, where μ is known.
- Prior: $\sigma^2\sim$ Scaled-inv-$\chi^2(\nu_0,\sigma_0^2)$
- Posterior: $\sigma^2\mid D\sim$ Scaled-inv-$\chi^2(\nu_n,s_n^2)$

$$\text{where }\nu_n=\nu_0+n,\quad s_n^2=\frac{\nu_0\sigma_0^2+\sum_{i=1}^n(X_i-\mu)^2}{\nu_0+n}$$

Posterior Predictive Distribution

Let $Y\sim N(\mu,\sigma^2)$ where μ is known. Find the marginal distribution $f(y\mid D)$:

$$\frac{Y-\mu}{s_n}\sim t_{\nu_n}\quad\textbf{(DO IT)}$$

Integration formula:

$$f(y\mid D)=\int f(y\mid\sigma^2)\,\pi(\sigma^2\mid D)\,d\sigma^2$$

Since $Y\mid\sigma^2\sim N(\mu,\sigma^2)$ and $\sigma^2\mid D\sim$ Scaled-inv-$\chi^2(\nu_n,s_n^2)$, the marginal distribution of Y is Student-t $(t_{\nu_n}(\mu,\sigma_0^2))$.

5. Unsolved / Exercise Problem

μ unknown & σ^2 follows non-informative prior:

$$p(\mu,\sigma^2)\propto\frac{1}{\sigma^2}\longrightarrow\text{Find }(\sigma^2\mid y)?$$

No Class
If any topic taught its in the W07 notes covered

Lec 10

Wednesday, August 19, 2026 2:03 PM

Multi-parameter models.

(θ_1, θ_2). $\pi(\theta_1, \theta_2)$ → Prior.

$X_1, \dots, X_n \mid \theta_1, \theta_2 \overset{iid}{\sim} f(x; \theta_1, \theta_2)$

e.g. $\theta_1 = \mu$, $\theta_2 = \sigma^2$.

$X_1, \dots, X_n \mid \mu, \sigma^2 \overset{iid}{\sim} N(\mu, \sigma^2)$

$\pi(\theta_1, \theta_2 \mid D) \propto \pi(\theta_1, \theta_2)\, \pi(D \mid \theta_1, \theta_2)$ ←, $\pi(\mu, \sigma^2)$ ✓.

μ, σ^2 → Nuisance parameters.

$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D)\, d\theta_2$.

(X, Y) $X^{(1)}, Y^{(1)}$ $X^{(2)}, Y^{(2)}$ $X^{(B)}, Y^{(B)}$

$\pi(X, Y) = \pi(X \mid Y)\pi(Y)$

$\pi(\theta_1 \mid D) = \int \pi(\theta_1, \theta_2 \mid D)\, d\theta_2$.

$= \int \pi(\theta_1 \mid \theta_2, D)\, p(\theta_2 \mid D)\, d\theta_2$ ←

$\pi(\theta_1 \mid \theta_2, D)$ $\pi(\theta_2 \mid D)$

Single-parameter inference.

$\pi(\theta_1 \mid D)$ Monte-Carlo integration

$\theta_1^{(1)}, \theta_2^{(1)}$
⋮
$\theta_1^{(B)}, \theta_2^{(B)}$

Marginal of $\theta_1 \mid D$ can be obtained by ignoring the samples of θ_2

$\theta_2 \to M_1, M_2$ → $\pi(\theta_1 \mid M_1, D)$

$\pi(\theta_2 \mid D)$

Suppose you want to estimate μ & σ^2 in $N(\mu, \sigma^2)$ model.

for known value of σ^2 you know how to estimate.

and you also know. $p(\sigma^2 \mid D)$

$\pi(\mu, \sigma^2 \mid D)$

$\pi(\mu \mid D) = \int \pi(\mu \mid \sigma^2, D)\, \pi(\sigma^2 \mid D)\, d\sigma^2$

σ^2 (known) — this needs to be calculated

Generating sample from $\pi(\mu \mid D)$ can be done in 2-steps:

Independent

Step1: Generate samples from $\sigma^2 \mid D$ (call it $\sigma^{2(s)}$).

Step2: $\mu \mid \sigma^{2(s)}, D$.

Gibbs sampling: $\sigma^2 \mid \mu, D \to \sigma^{(s)}$ → conditional.

$\mu \mid \sigma^2, D \to \mu^{(s)}$ → conditional

→ Dependence.

$\sigma^{2(s+1)} \mid \mu^{(s)}, D$

$\mu^{(s+1)} \mid \sigma^{2(s+1)}, D)$

in Gibbs sampler.

Go to step 1 & Iterate

(X, Y) $\pi(X)$ → Generate from marginal dist^n of X

& then $\pi(Y \mid X)$ → Conditional dist^n

Repeat

Scaled-inverted-χ^2 distribution:

$$f(x; \nu, \tau^2) = \tau^{\nu} \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \frac{1}{x^{(\frac{\nu}{2}+1)}} e^{-\frac{\nu\tau^2}{2x}}, \quad x > 0$$

Normal unknown σ^2,

Scaled-inv-$\chi^2(\nu, \tau^2)$. ν is degrees of freedom. τ^2 is scale

Known μ.

If $X \sim$ Scaled-inv-$\chi^2(\nu, \tau^2)$.

then a) $\frac{\nu\tau^2}{X} \sim \chi^2(\nu)$ b) $X \sim$ inv-Gamma$\left(\frac{\nu}{2}, \frac{\nu\tau^2}{2}\right)$

c) $kX \sim$ Scaled-inv-$\chi^2(\nu, k\tau^2)$

$\left[\frac{\nu\tau^2 k}{kX} \sim \chi^2(\nu) \Rightarrow kX \sim \text{Scaled-inv-}\chi^2(\nu, k\tau^2)\right]$

Let $X_1, \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ μ is known.

Let $\sigma^2 \sim$ Scaled-inverse-$\chi^2(\nu_0, \sigma_0^2)$.

Then, $\sigma^2 | D \sim$ Scaled-inverse-$\chi^2(\nu, s^2)$

where $\nu = \nu_0 + n$, $s^2 = \frac{\nu_0\sigma_0^2 + \Sigma(x_i - \mu)^2}{\nu_0 + n}$

$\sigma^2 \sim$ Scaled-inverse-$\chi^2(\nu_0, \sigma_0^2)$.

$Y | \sigma^2 \sim N(\mu, \sigma^2)$ μ is known

Then, find the marginal dist^n of Y.

$Y \sim t_{\nu_0}(\mu, \sigma_0^2)$ → DO IT

$\sigma^2 \rightarrow$ known; $\mu \rightarrow$ unknown & μ follows normal

↑ Marginal of Y is normal.

$p(y) = \int p(y|\sigma^2)\pi(\sigma^2)\, d\sigma^2$

$y|\sigma^2 \sim N(\mu, \sigma^2)$

$\sigma^2 | D \sim$ Scaled-inv-$\chi^2(\nu_n, s_n^2)$

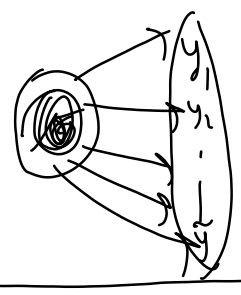
Q $p(\tilde{y}|D)$ (Posterior predictive distribution).

D: $y_1 \ldots y_n \overset{iid}{\sim} N(\mu, \sigma^2)$ μ known.

$\sigma^2 \sim$ Scaled-inv-$\chi^2(\nu_0, \sigma_0^2)$

$\sigma^2 | D \sim$ Scaled-inv-$\chi^2(\nu_n, s_n^2)$

or $\tilde{y} | \sigma, D$)



$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$ → Find.

y: $y_1, \ldots y_n \overset{iid}{\sim} N(\mu, \sigma^2)$

Scaled-inverse-χ^2 → $p(\sigma^2|y)$?

Lec 11

Tuesday, September 1, 2026 2:43 PM

Unknown Mean & Unknown Variance.

Assume prior as: $p(\mu,\sigma^2) \propto \frac{1}{\sigma^2}$

$y_1, y_2 \ldots y_n \overset{iid}{\sim} N(\mu,\sigma^2)$.

y

$p(\mu,\sigma^2|y)$.

Independent Jeffreys prior for μ & σ^2

$p(\theta_1,\theta_2|y)$

$\propto p(\theta_1|\theta_2,y)\, p(\theta_2|y)$.

When variance is known you can do this calculation.

$$p(\mu,\sigma^2|y) \propto \sigma^{-n-2} \exp\left\{-\frac{\sum(y_i-\mu)^2}{2\sigma^2}\right\}$$

$p(\mu|\sigma^2,y)$ $\quad$ $p(\sigma^2|y)$

$p(\mu|\sigma^2,y)$ → (A)

When $\mu \sim N(\mu_0,\tau_0^2)$ & σ^2 is known.

Then, $p(\mu|\sigma^2,y) \sim N(\mu_1,\tau_1^2)$ where $\mu_1 = \left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{n}{\sigma^2}\right)\bar{y}$ / $\left(\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}\right)$

$\pi(\mu) \propto 1$. can be obtained by taking $\tau_0 \to \infty$ in (A) & $\tau_1^2 = \frac{1}{\frac{n}{\sigma^2}+\frac{1}{\tau_0^2}}$

$$\mu|\sigma^2,y \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right) \quad \text{(R1)}$$

$$p(\sigma^2|y) \propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}\left\{\sum\frac{(y_i-\mu)^2}{2\sigma^2}\right\}\right] d\mu$$

$\sum(y_i-\bar{y})^2 + n(\bar{y}-\mu)^2$

$\propto \sigma^{-(n+1)} \exp\left\{-\frac{(n-1)s^2}{2\sigma^2}\right\}$ where $s^2 = \frac{\sum(y_i-\bar{y})^2}{n-1}$

$$\sigma^2|y \sim \text{Scaled-Inverse-}\chi^2(n-1,s^2) \quad \text{(R2)}$$

Joint posterior

$p(\mu,\sigma^2|y) \propto$ R1 X R2

$p(\mu,\sigma^2|y)$

$p(\sigma^2|y) \times p(\mu|\sigma^2,y)$

Sample from joint posterior.

Sample from $\sigma^2|y$ &

then sample $p(\mu|\sigma,\cdot)$ ... for sample 1 =.

Classical inference.

(R2) $\Rightarrow$ $\frac{(n-1)s^2}{\sigma^2}\,|\,y \sim \chi^2_{(n-1)}$ $\longleftrightarrow$ $\frac{(n-1)s^2}{\sigma^2}\,|\,\mu \sim \chi^2_{(n-1)}$

Pivotal ~~statistics~~ quantity.

~~$N(\mu,1)$~~ $N(\mu,1)$. $N(0,1)+\mu$

location family:

$p(y-\mu)$ is independent of μ. pdf of $(y-\mu)|\mu$ is independ of location family.

y is generated from a.

$\Rightarrow$ $u = y-\mu$ and let $g(u)$ is the pdf of u. μ.

Then. ~~$p(y|\mu)$~~ $p(y|\mu) = g(u)\left|\frac{du}{dy}\right|$

$= g(u) = g(y-\mu)$

(Kind of) Ancillary for μ

$y-\mu$ → y, (μ) ; y & (μ) independent

$p(y-\mu|y) = g(u)$

~~$u = y-\mu$~~ $p(u|y) = g(u)$

$p(\mu|y) = g(u)\left|\frac{du}{d\mu}\right|$.

$= g(u)$

Reference prior

Pivotal quantity

~~$p(\mu|y) \propto \pi(\mu)\, p(y|\mu)$~~.

$g(u) = \pi(\mu)\, g(u)$ ⁂

$\Rightarrow \pi(\mu) \propto 1$

$p\left(\frac{y}{\theta}\,|\,\theta\right)$ is a function that is free of θ

$\frac{y}{\theta}$ e.g. $z \sim \text{Exp}(1)$ & then $y = z\theta$.

$\frac{y}{\theta}$ is a pivotal quantity here. & θ is called a ~~for~~ scale parameter.

$u = \frac{y}{\theta}$.

$p\left(\frac{y}{\theta}\,|\,\theta\right)$ should be free of θ

$u = \Delta u$

$\Rightarrow p(u|\theta) = \cdots$

$p(y|\theta) = g(u)\left|\frac{du}{dy}\right|$

$= \frac{g(u)}{\theta} = \frac{g(y/\theta)}{\theta}$

$\frac{y}{\theta}$ y

dist^n of y depend on θ & dist^n of $\frac{y}{\theta}$ doesn't depend on θ.

$\frac{y}{\theta}$ & y should be independent.

$p\left(\frac{y}{\theta}|y\right) = g(u)$

$u = \frac{y}{\theta} \Rightarrow \theta = \frac{y}{u} \Rightarrow \left|\frac{du}{d\theta}\right| \propto \frac{1}{\theta^2}$

$p(\theta|y) = p(u|y)\left|\frac{du}{d\theta}\right| = \frac{g(u)}{\theta^2}$

$p(\theta|y) \propto \pi(\theta)p(y|\theta)$

$\Rightarrow \frac{g(u)}{\theta^2} \propto \frac{\pi(\theta)}{\theta} g(u) \Rightarrow \pi(\theta) \propto \frac{1}{\theta}$ Reference prior

---

Conjugate prior for unknown mean μ & unknown variance σ^2.

$p(y|\mu,\sigma^2) \propto \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left\{-\frac{\sum(y_i-\mu)^2}{2\sigma^2}\right\}$

$\propto (\sigma^2)^{-n/2} \exp\left\{-\frac{\sum(y_i-\bar{y})^2 + n(\bar{y}-\mu)^2}{2\sigma^2}\right\}$

$= (\sigma^2)^{-n/2} \exp\left\{-\frac{(n-1)s^2}{2\sigma^2}\right\} \exp\left\{-\frac{n(\bar{y}-\mu)^2}{2\sigma^2}\right\}$

Conjugate prior

Scaled-Inverse-χ^2

$p(\mu,\sigma^2) \propto p(\sigma^2)\, p(\mu|\sigma^2)$

$(\mu|\sigma^2) \sim N(\,)$

where $p(\sigma^2)$ is Scaled-Inverse-χ^2 and $\mu|\sigma^2 \sim N(\,)$

$\mu|\sigma^2 \sim N\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right)$

$\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$

$\Rightarrow p(\mu,\sigma^2) = N\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right)$ Scaled-Inv-$\chi^2(\,)$

$\Rightarrow p(\mu) \quad \mu \sim t_{\nu_0}\left(\mu_0, \frac{\sigma_0}{\kappa_0}\right)$

$p(\mu,\sigma^2|y) = N\text{-inv-}\chi^2\left(\mu,\sigma^2 \,\middle|\, \mu_n, \frac{\sigma_n^2}{\kappa_n}, \nu_n, \sigma_n^2\right)$

where $\mu_n = \frac{\kappa_0}{\kappa_0+n}\mu_0 + \frac{n}{\kappa_0+n}\bar{y}$, $\kappa_n = \kappa_0+n$, $\nu_n = \nu_0+n$, $\nu_n\sigma_n^2 = \nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_0+n}[\bar{y}-\mu_0]^2$

$p(\mu|y)$ $\quad p(\tilde{y}|y)$

① Show the dist^n of $\mu|y \sim t_{n-1}\left(\bar{y}, \frac{s^2}{n}\right)$ when prior $p(\mu,\sigma) \propto \frac{1}{\sigma^2}$

② Find the dist^n of $\tilde{y}|y$ ($\tilde{y}$ is a new obs.) using properties of normal & scaled-Inverse-χ^2.

Wednesday, September 2, 2026 2:28 PM

MAP & Credible Intervals

$\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}}\, L(\theta; x)$ — x: Data; θ: parameter

Data is given, $L(\theta; x) = f(\theta|x)$

$\phi = h(\theta)$ h: one-one.

$\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$

MLE is invariant for 1-1 transformation

i.e. $\hat{\beta}_{MLE} = h(\hat{\theta}_{MLE})$

Maximum Aposteriori Estimation. (MAP)

$\pi(\theta|x)$

MAP is a point estimator of θ based on $\pi(\theta|x)$

$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}}\, \pi(\theta|x)$

: mode of $\pi(\theta|x)$.

$X \sim N(\mu, \sigma^2)$. mode is μ

$Y = e^X$, $X = \log Y$ e^{μ} (is the mode?)

$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$

$f_Y(y) =$

$\therefore$ Mode of Y will be $e^{\mu - \sigma^2} \neq e^{\mu}$

$L(\theta; x) \Rightarrow p(x|\theta)$ $\pi(\theta|x)$

$L(\theta; x)$ (Likelihood function is not a prob. density fn.)

$\int L(\theta; x)\, d\theta \neq 1$ $\int p(x|\theta)\, dx = 1$

$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}}\, \pi(\theta|x)$ $\pi(\theta|x)$ is a prob. density fn.

$= \underset{\theta}{\operatorname{argmax}} \left(L(\theta; x) \cdot \pi(\theta) \right)$ therefore MAP is not invariant to reparameterization.

Prior

e.g. $X_1, \ldots X_n \overset{iid}{\sim} \exp(\beta)$.

$\pi(\beta) \propto 1$. Then $\hat{\beta}_{MAP} = \frac{\sum x_i}{n}$.

$pdf = \frac{1}{\beta} \exp(-\frac{x}{\beta})$. $\hat{\beta}_{MLE} = \frac{\sum x_i}{n}$

$\pi(\beta|x) \propto \frac{1}{\beta^n} \exp(-\frac{\sum x_i}{\beta}) \Rightarrow \pi(\phi|x)$

$\phi(\beta) = \log \beta$.

& then find $\hat{\phi}_{MAP}$; $\hat{\phi}_{MLE}$

$\hat{\phi}_{MLE} = \log(\frac{\sum x_i}{n})$

$\log\left(\frac{\sum x_i}{n-1}\right)$

Posterior mean $\pi(\phi|x)$.

mode (MAP)

Credible interval.

$P(L < \theta < U|x) = (1-\alpha)$

$P(L < \theta < U) = 1 - \alpha$

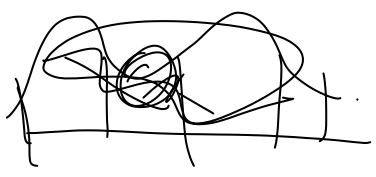


e.g. $X|p \sim \text{Binomial}(n, p)$.

$p \sim \text{Beta}(a, b)$

$p|X \sim \text{Beta}(a+x, b+n-x)$

$\left(F^{-1}_{\text{Beta}(a+x, b+n-x)}(0.025),\ F^{-1}_{(a+x, b+n-x)}(0.975) \right)$



$X_1, \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. σ known.

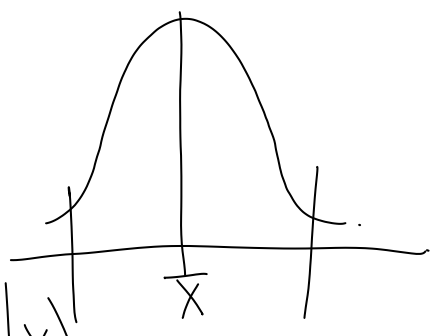
$\pi(\mu) \propto 1$.

Then, $\pi(\mu|x)$ $N(\bar{X}, \frac{\sigma^2}{n})$

$\mu|X_1 \ldots X_n \sim N(\bar{X}, \frac{\sigma^2}{n})$

$\exp -\frac{\sum (x_i - \mu)^2}{2\sigma^2}$ $\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$

$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}} \right)$.



$Pr\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}} \,\middle|\, X \right) = 0.95$

$\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) \sim N(0, I^{-1}(\theta))$

$\left(\hat{\theta} - \frac{1.96\sigma}{\sqrt{n}}, \hat{\theta} + \frac{1.96\sigma}{\sqrt{n}}\right)$ Asymptotic normality of MLE

Bernstein von Mises theorem $\sqrt{n}(\theta - \theta_0)\,|\,\mathbf{X} \sim N(0, \sigma_0)$

(1) Paradox of classical inference.

(2) Why Bayesian. How to do Bayesian (prior, likelihood $\Rightarrow$ posterior)

(3) Prior → Conjugate prior (good for calculations)

Some concepts of non informative prior. (Jeffreys).

prior, prior predictive dist^n, posterior, posterior predictive dist^n, updating belief

→ Check atleast all the cases mentioned in BSDS syllabus.

Properties of scaled-inverse-χ^2 dist^n

→ unknown mean unknown variance & Dirichlet Multinomial

MAP & credible interval

Wednesday, September 2, 2026 2:28 PM

MAP & Credible Intervals

$\hat{\theta}_{MLE} = \arg\max_{\theta} L(\theta; x)$ — θ: parameter, x: Data.

Data is given, $L(\theta; x) = f(\theta|x)$

$\phi = h(\theta)$ h: one-one.

$\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$

MLE is invariant for 1-1 transformation
i.e. $\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$

Maximum A posteriori Estimation. (MAP)

$\pi(\theta|x)$

MAP is a point estimator of θ based on $\pi(\theta|x)$

$MAP = \arg\max_{\theta} \pi(\theta|x)$
: mode of $\pi(\theta|x)$.

$X \sim N(\mu, \sigma^2)$. mode is μ

$Y = e^X$, $X = \log Y$

e^{μ} (is the mode?)

$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$

$f_Y(y) =$

! Mode of Y will be $e^{\mu - \sigma^2} \neq e^{\mu}$

$L(\theta; x) \Rightarrow p(x|\theta)$

$\pi(\theta|x)$

$L(\theta; x)$ (Likelihood function is not a prob. density fn.)

$\int L(\theta; x) d\theta \neq 1$ $\left[\int p(x|\theta) dx = 1 \right]$

$\hat{\theta}_{MAP} = \arg\max_{\theta} \pi(\theta|x)$

$= \arg\max_{\theta} \{ L(\theta; x) \cdot \pi(\theta) \}$ ← Prior

$\pi(\theta|x)$ is a prob. density fn. therefore MAP is not invariant to reparameterization.

e.g. $X_1, \dots X_n \overset{iid}{\sim} \exp(\beta)$.

$\pi(\beta) \propto 1$. Then $\hat{\beta}_{MAP} = \frac{\sum x}{n}$

pdf $= \frac{1}{\beta} \exp(-\frac{x}{\beta})$. $\hat{\beta}_{MLE} = \frac{\sum X_i}{n}$

$\pi(\beta|x) \propto \frac{1}{\beta^n} \exp(-\frac{\sum X_i}{\beta}) \Rightarrow \pi(\phi|x)$

$\phi = \log \beta$.

& then find $\hat{\phi}_{MAP}$; $\hat{\phi}_{MLE}$

$\hat{\phi}_{MLE} = \log(\frac{\sum X_i}{n})$

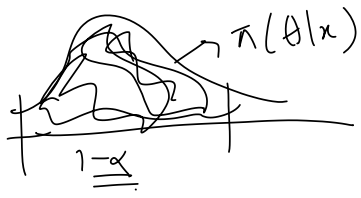
$\log(\frac{\sum X_i}{n-1})$

Posterior mean $\pi(\phi|x)$.
mode (MAP)

Credible interval

$P(L \leq \theta \leq U | x) = (1-\alpha)$

$P(L < \theta < U) = 1 - \alpha$

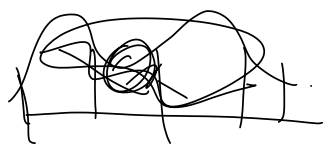


e.g. $X|p \sim \text{Binomial}(n, p)$.

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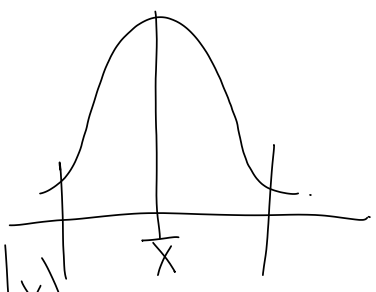
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$\mu | X_1 \dots X_n \sim N(\bar{X}, \frac{\sigma^2}{n})$

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Bernstein von Mises theorem

$\sqrt{n}(\theta - \theta_0) | X \rightsquigarrow N(0, \sigma_0)$

(1) Paradox of classical inference.
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updating belief
→ Check atleast all the cases mentioned in BSDS Syllabus.
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→ unknown mean unknown variance & Dirichlet Multinomial
MAP & Credible interval

Lec 12 (After Midsem)

Wednesday, September 16, 2026 2:11 PM

Estimation : Prior over the parameter · × Likelihood · → $P(\theta|X)$ ✓ → MAP, Credible interval.

Bayesian hypothesis testing: Posterior odds, Bayes Factor

Diagnostic example : $P(D) = 0.02$, $P(\bar{D}) = 0.98$; $P(T^+|D) = 0.95$, $P(T^+|\bar{D}) = 0.10$

$$P(D|T^+) = \frac{P(D)\,P(T^+|D)}{P(T^+)} \simeq 0.1624$$

Posterior $(P(D|T^+)) \propto P(T^+|D)\,P(D)$

Hypothesis testing setup: $H_0 : \bar{D}$ $H_1 : D$

$P(H_0) = 0.98$ · $P(H_1) = 0.02$

· Compute posterior probabilities of both the hypotheses

Accept H_1 if $P(H_1|x) > 0.5$ [$P(\bar{D}|T^+) > 0.5$] [Decision Rule]

if $\frac{P(H_1|x)}{P(H_0|x)} > 1$.

then accept H_1.

Posterior odds : $\frac{P(H_1|x)}{P(H_0|x)} = \frac{P(x|H_1)}{P(x|H_0)} \cdot \frac{P(H_1)}{P(H_0)}$

Posterior odds · Likelihood ratio · Prior odds

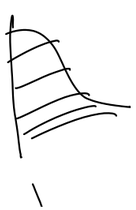
Bayes Factor : $BF_{10} = \frac{P(x|H_1)}{P(x|H_0)}$ & then, $\frac{P(H_1|x)}{P(H_0|x)} = BF_{10}\,\frac{P(H_1)}{P(H_0)}$

BF_{10} : Compatibility of the observed data with the hypotheses.

$BF_{10} = \frac{P(T^+|D)}{P(T^+|\bar{D})} = 9.5$

Posterior odds = BF_{10} (prior odds)

Bayes Factor doesn't depend on the prior odds of the hypotheses.





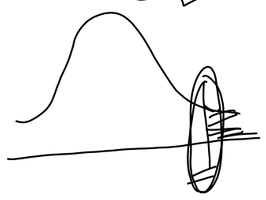
Composite hypotheses:

Suppose $X|\theta, H_j \sim f_j(x|\theta)$.

under H_j, suppose $\pi_j(\theta)$ → Prior for θ.

$H_0: \mu = 0$ vs $H_1: \mu \neq 0$

$H_0: \mu \leq 0$ vs $H_1: \mu > 0$



~~$H_0: \mu = 0$ vs $H_1: \mu$~~

$H_0: \mu \geqslant 0$ vs $H_1: \mu > 0$

$\pi_1(\mu)$ $\pi_2(\mu)$

$$P(x|H_j) = \int f_j(x|\theta)\,\pi_j(\theta)\,d\theta$$

(10) → $H_0: X \sim N(\mu, 1)$, $\mu \sim U(9, 11)$

$m_j(x) =$ (Marginal likelihood)

(Prior predictive ~~dis~~ density)

$f(10; 9, 1)$ $H_1: X \sim N(\mu, 1)$, $\mu \sim N(-9, 1)$

$\mu = U[-11, -9]$

$f(10; \mu, 1)$

$f(10; -11, 1)$

$f(10; -9, 1)$

$$BF_{10} = \frac{m_1(x)}{m_0(x)}$$ → Not using prior odds of the hypotheses.

$$\frac{P(H_1|x)}{P(H_0|x)} \propto \frac{P(x|H_1)}{P(x|H_0)} \frac{P(H_1)}{P(H_0)}$$ → Prior odds.

When prior odds are same,

posterior odds = BF_{10}

$H_1: X \sim N(\mu, 1)$, $\mu \sim N(9, 1)$

$H_0: X \sim N(\mu, 1)$, $\mu \sim N(-9, 1)$

D: 10

$BF_{10} > 1$

Rule of thumb

if $P(H_1|x) > P(H_0|x)$ then accept H_1 // accept H_0

evidence = f(marginal likelihood)

if $\log_{10}(BF_{10}) \in (0, 0.5)$ the evidence ~~against~~ H_0 is

Likelihood (θ_1) / Likelihood (θ_2)

marginal likelihood (H_1) / marginal likelihood (H_0)

$H_0: \theta = \theta_1$

$H_1: \theta = \theta_2$

$\frac{m_1(x)}{m_0(x)}$ then BF = likelihood ratio

$\mathcal{E}(0.5,1)$ — poor

$\mathcal{E}(1,2)$ — the evidence against H_0 is substantial.

$\mathcal{E}(>2)$ — strong / decisive

$H_0: \mu = 0$ $H_1: \mu \neq 0$

$\mu = 0$ w.p.1 $\mu \sim N(1,1)$

Note that,

Likelihood: $\dfrac{f(x|\theta_1)}{f(x|\theta_2)}$

Likelihood ratio

$\int f(x|\theta_1)\,dx = 1$

$\int f(x|\theta_1)\,dx = 1$

$\int f(x|\theta_2)\,dx = 1$

Show this $\int m_1(x)\,dx = 1$

$\int f(x|\theta)\,d\theta \neq 1$

$\int f(x|\theta)\,dx = 1$

Lec 14

Tuesday, September 22, 2026 2:38 PM

M_1 and M_2

H_0: Data is coming from M_1 ($f(x,\theta_1)$ & $\pi(\theta_1)$)

H_1: ——— M_2 ($f(x,\theta_2)$ & $\pi(\theta_2)$)

Proper priors for θ_1 & θ_2 i.e. $\pi(\theta_1)$, $\pi(\theta_2)$ are pdf.

marginal likelihood M_j = $\int f(\text{data}|\theta_j)\pi(\theta_j)\,d\theta_j$

$P(M_1)$, $P(M_2)$

$P(M_j|\text{data})$ $j=1,2$

BF → calculates evidence of the comparability of the data for both the models.

Data: y

$$BF = \frac{\pi(M_1|y)}{\pi(M_1)} \Big/ \frac{\pi(M_2|y)}{\pi(M_2)} = \frac{m(y|M_1)}{m(y|M_2)}$$

Example: H_0: $\theta=\theta_0$ vs H_1: $\theta\neq\theta_0$ density function is $f(x|\theta)$

$X=x$

under H_0: the distribution of θ is degenerate at θ_0 (Null hypothesis simple)

under H_1: $\theta\sim g_1(\theta)$ prior of θ under H_1.

Let the parameter space for θ is continuum (i.e. $\theta\in\mathbb{R}$)

We need to find the marginal likelihood under both H_0 and H_1.

Marginal likelihood under H_1: $m_1(x) = \int f(x|\theta)\,g_1(\theta)\,d\theta$

And $P(H_0)=p_0$ & $\pi(H_1)=1-p_0$

$\pi(\theta)=p_0 g_1(\theta) + (1-p_0)g_2(\theta)$ | $P(H_0)=p_0$

$=p_0 I_{\{\theta=\theta_0\}} + (1-p_0)g_2(\theta)$

$$\pi(\theta=\theta_0|x) = \frac{p_0 f(x|\theta_0)}{p_0 f(x|\theta_0)+(1-p_0)m_1(x)}$$ → marginal dist^n for H_1.

$\pi(H_0|x)$

$$= \left[1+\left(\frac{1-p_0}{p_0}\right)\frac{m_1(x)}{f(x|\theta_0)}\right]^{-1} = \left[1+\frac{1-p_0}{p_0}\frac{1}{B_{01}(x)}\right]^{-1}$$

where $B_{01}(x) = \frac{f(x|\theta_0)}{m_1(x)}$

Example: Let $X\sim \text{Binomial}(n,p)$.

and consider testing H_0: $p=1/2$ vs H_1: $p\neq 1/2$, $g_1(p)=1$.

$n=5$, $x=3$ & then what will be the Bayes factor.

Ans: $\frac{15}{8}$

under H_0: Likelihood is $\binom{n}{x}\left(\frac{1}{2}\right)^n$ (marginal)

$P(H_0)=P(H_1)=1$.

$P(H_0|x)=$

Ex: H_0: $p<0.5$ vs H_1: $p\geq 0.5$

$X|p\sim \text{Binomial}(n,p)$.

→ $\pi_0(p)=2$, $0<p<0.5$

$\pi_1(p)=2$, $0.5<p<1$

$n=10$ & $X=7$

$2\binom{n}{x}\int_0^{0.5} p^x(1-p)^{n-x}\,dp$: H_0

$\int_{0.5}^{1} p^x(1-p)^{n-x}\,dp$: H_1

Ex: H_0: $X_i \overset{iid}{\sim} N(0,1)$

H_1: $X_i \overset{iid}{\sim} \text{Cauchy}(0,1)$ $\left(f(x|0)=\frac{1}{\pi(1+x^2)}\right)$

$x=(0.2,-0.4,0.7,-0.1,0.3)$

$BF_{10}\approx 0.43$

Posterior odds / Prior odds = Bayes factor

$P(H_0)$ & $P(H_1)$

Ex: H_0: $X_i \overset{iid}{\sim} N(\mu,\sigma^2)$ $(\pi_1(\mu,\sigma^2))$

H_1: $X_i \overset{iid}{\sim} (\text{anything})(\mu,\sigma)$ $(\pi_2(\mu,\sigma^2))$

$$BF_{10} = \frac{\int L_1(\mu,\sigma|x)\pi_1(\mu,\sigma)\,d\mu\,d\sigma}{\int L_2(\mu,\sigma|x)\pi_2(\mu,\sigma)\,d\mu\,d\sigma}$$

Ex: $X_1,\dots X_n|\mu \overset{iid}{\sim} N(\mu,1)$.

H_0: $\mu\sim N(0,1)$

H_1: $\begin{cases} X_i|\mu\sim N(\mu,1) \\ \mu\sim N(0,1) \end{cases}$ ⟹ Connected by μ

H_0: $\mu \sim N(0,100)$

$n=10$, $\bar{x}=0.5$

$BF_{10} = \frac{\phi(0.5; 0, 100.1)}{\phi(0.5; 0, 1.1)} \approx (0.11)$

Occam's Razor (You pay a price for the flexibility of the model)

H_1: $(X_i|\mu \sim N(\mu,1)$, $\mu \sim N(0,100))$ — Prior factor

$\bar{X}|\mu \sim N(\mu, \frac{1}{n})$, $\mu \sim N(0, \tau^2)$

$\Rightarrow \bar{X}|H_1 \sim N(0, \tau^2 + \frac{1}{n})$

$\therefore \bar{X}|H_1 \sim N(0, 100 + \frac{1}{n})$

$\bar{X}|H_0 \sim N(0, 100 + \frac{1}{n})$

$\Sigma(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$

Bayes Factor consistency: Can we select the correct hypothesis using Bayes Factor?

Ex: $X_1 \dots X_n | \mu \overset{iid}{\sim} N(\mu, 1)$

$H_0: \mu = 0$ vs $H_1: \mu \neq 0$ ($\mu \sim N(0, \tau^2)$)

Consistency: if $n \to \infty$, your inference should be correct

$\mu \leq 0$ $\quad \mu > 0$ $\rightarrow$

$N(0,\tau^2) 1_{(-\infty,0)}$ $\quad N(0,\tau^2) 1_{(0,\infty)}$

if $X_1 \dots X_n \overset{iid}{\sim} N(0,1)$ then $BF_{10,n} \to \begin{cases} 0 & \\ \text{o/w } BF_{10,n} \to \infty & \end{cases}$ (0.9 a.s.)

$\rightarrow \mu = 0 \Rightarrow BF_{10,n} \to 0$ a.s.

$\mu \neq 0 \Rightarrow BF_{10,n} \to \infty$ a.s.

$BF_{10,n} = f(X_1, \dots, X_n)$

$\bar{X}|H_0 \sim N(0, \frac{1}{n})$ $\quad \bar{X}|H_1 \sim N(0, \tau^2 + \frac{1}{n})$

$BF_{10,n} = \frac{\phi(\bar{X}; 0, \tau^2 + 1/n)}{\phi(\bar{X}; 0, \frac{1}{n})} \Rightarrow \log BF_{10,n} = \frac{-1}{2}\log(1+n\tau^2) + \frac{\bar{X}^2 n^2 \tau^2}{2(n\tau^2+1)}$

$X_1 \dots X_n \overset{iid}{\sim} N(0,1) \Rightarrow \sqrt{n}\bar{X} \sim N(0,1)$

$\Rightarrow n\bar{X}^2 \sim \chi^2_{(1)}$

$\left(n\bar{X}^2\right)\left(\frac{n\tau^2}{n\tau^2+1}\right)\frac{1}{2} \approx \frac{n\bar{X}^2}{2} - \log(n)$

$\therefore \mu = 0 \Rightarrow BF_{10,n} \to 0$ a.s.

$\Rightarrow P(H_0|X_1 \dots X_n) \to 1$ if $\mu = 0$ in the true data-generating process. a.s.

$\to -\infty$ a.s.

$n\bar{X}^2 \to \chi^2_{(1)}$

$\log BF_{10,n} = \frac{-1}{2}\log(1+n\tau^2) + \frac{\bar{X}^2 n^2 \tau^2}{2(n\tau^2+1)}$ $\quad \bar{X}$

Let, $X_1 \dots X_n \overset{iid}{\sim} N(\mu^*, 1)$ & $\mu^* \neq 0$ $\quad \bar{X} \to \mu^* \Rightarrow \bar{X}^2 \to (\mu^*)^2$

$\therefore \log BF_{10,n} \xrightarrow{p} \infty$ $\quad n\bar{X}^2 \to \infty$

$\therefore P(H_1|X_1 \ldots X_n) \xrightarrow{P} 1$

$H_0: \mu = 0$ vs $H_1: \mu = 1$ But $\mu \in \{0, 1\}$

$P(H_0|\text{Data}) + P(H_1|\text{Data}) = 1$

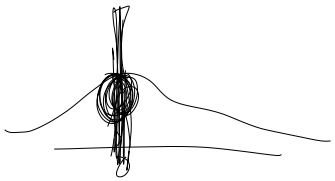
Ex $X_1 \ldots X_n \overset{iid}{\sim} N(\mu, 1)$

$H_0: \mu \sim N(9,1) \quad H_1: \mu \sim N(-9,1)$

Let the data-generating process be $X_i \overset{iid}{\sim} N(10,1)$

$$BF_{01,n} = \frac{\phi(\bar{X}_n; 9, 1+\frac{1}{n})}{\phi(\bar{X}_n; -9, 1+\frac{1}{n})} = \exp\left\{\frac{18\bar{X}_n}{1+\frac{1}{n}}\right\} \to \exp\{180\}$$

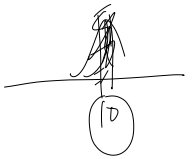
$\therefore P(H_0|X_1 \ldots X_n) \to 1$ as $n \to \infty$



Ex: $H_0: \mu \le 0 \quad H_1: \mu > 0$

$\mu \sim TN(0,1)_{(-\infty,0]} \quad \mu \sim TN(0,1)_{(0,\infty)}$

Data generating process: $X_1 \ldots X_n \overset{iid}{\sim} N(10,1)$



Solve it

Ex: $H_0: \mu \le 0 \quad H_1: \mu > 0$

$\mu \sim TN(0,1)_{(-\infty,0]} \quad \mu \sim TN(0,1)_{(0,\infty)}$

$X_1 \ldots X_n \overset{iid}{\sim} N(0,1)$

Simulation (3 marks)

Write the details of simulation (3 marks)

14/11/2026 11.59 pm
assignment.cda@gmail.com.

Q2 (2 marks)

a) Show that as $n \to \infty$ $\sqrt{n}\left(\theta - \hat{\theta}_T \mid X_1 \ldots X_n\right) \to N(0, \Sigma)$

generate $X_1 \ldots X_n \overset{iid}{\sim}$ Double Exponential (θ).

b) Same as (a) only now $\sqrt{n}\left(\theta^2 - \theta_T^2 \mid X_1 \ldots X_n\right) \to N(0, \Sigma)$

Write the code in rstan package

Q3 (3 marks)

$X_1 \ldots X_n \overset{iid}{\sim} pN(\mu_1, 1) + (1-p)N(\mu_2, 1)$

True values $p = 0.2$, $\mu_1 = 0$, $\mu_2 = 4$

Then using simulation find the ~~dist~~ posterior dist^n of p, μ_1, μ_2

$\pi(\mu_1) = \pi(\mu_2)$

$0.2\,N(0,1) + 0.8\,N(4,1)$

$0.8\,N(4,1) + 0.2\,N(0,1)$ ~~$p = 0.8$ $\mu_1 = 4$ $\mu_2 = 0$~~

---

(1) Markov Chain
(2) Gibbs sampling
(3) Metropolis - Hastings

Diagnostics of convergence

Linear Model
Hierarchical

Data analysis

$f(x, y, z)$

$\frac{f(x, y, z)}{\pi(\theta, \gamma)}$, $\pi(\theta)\, b(\gamma \mid \theta)$

Tuesday, September 29, 2026 2:29 PM

Linear Regression

↳ (Bayesian inference for linear regression.)

$Y_1, \ldots Y_n \overset{iid}{\sim} N(\mu, \sigma^2)$ — univariate.

$Y_1, \ldots Y_n \to Y_i \overset{ind}{\sim} N(x_i^T \beta, \sigma^2)$ — multivariate

MAP

Credible interval

$\mu \to \pi(\mu) \propto 1$.

$\mu \sim N(\mu_0, \tau^2)$.

$\beta_j = 0$ vs $\beta_j \neq 0$

Find credible interval.

$\beta \to \pi(\beta) \propto 1$.

$\beta \sim N(\beta_0, \tau_0^2)$

Prior ~~distribution~~

Prior ≠ distⁿ.

Prior predictive distⁿ.

Posterior predictive distⁿ.

LASSO / RIDGE

Bayesian formulation. equivalent to LASSO RIDGE.

Missingness → ~~Removed~~ Parameter.

$Y = X\beta + \varepsilon$. $Y = (Y_1, \ldots Y_n)^T$.

$\beta = (\beta_0, \beta_1, \ldots \beta_p)^T$.

$\varepsilon \sim N(0, \sigma^2 I_n)$

Probabilities are essential for Bayesian inference. ∴ We need to have some distⁿ for ε as well

$Y|X$ follows some distⁿ (conditional distⁿ. (Regression)).

~~Q.~~ Does the distⁿ of X matter?

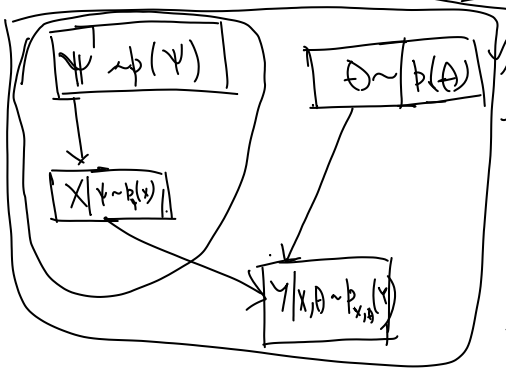
$\ldots \mid y, x) \propto \ldots$ in the ~~to~~ regression model.

X $p(X|\Psi)$ → parameter. and then $p(\Psi|\theta)$ ... with unknown parameter θ.



$p(\Psi, \theta | X, Y)$

Prior: $p(\Psi, \theta) = p(\Psi)\, p(\theta)$



$p(\Psi, \theta, X, Y)$
$= p(\Psi)\, p(\theta)\, p(X|\Psi)\, p(Y|X,\theta)$

$\Rightarrow \dfrac{p(\Psi, \theta, X, Y)}{p(X, Y)}$

$p(\Psi|X,Y) = \dfrac{p(\Psi, X, Y)}{p(X, Y)}$

$= \dfrac{\int p(\Psi, \theta, X, Y)\, d\theta}{p(X, Y)}$

$\propto p(\Psi)\, p(X|\Psi)$

$p(\theta|X,Y) \propto p(\theta)\, p(Y|X,\theta)$

Data: X_1, Y_1 ... X_n, Y_n & Prior for θ.

$Y|X,\theta \sim \text{dist}^n(X,\theta)$

Show that

$p(\Psi, \theta | X, Y) \propto p(Y|X)\, p(\theta|X,Y)$

Prior predictive dist^n.

$x^* = \begin{pmatrix} 1 \\ x_1^* \end{pmatrix}$

$Y^*|\beta, x^* \sim N(x^{*T}\beta, 1)$ (β, Σ are independent)

$\beta \sim N(0, \tau^2 I)$

What is the dist^n of $Y^*|x^*$

$Y^* = X^{*T}\beta + \varepsilon$ $\quad E(Y^*|X^*) = E(x^{*T}\beta + \varepsilon)$

$Y^*|X^*$ $\quad = 0$

$V(x^{*T}\beta + \varepsilon | X^*) = \tau^2 x^{*T} x^* + 1$.

τ : known variance of β.

$\beta = (\beta_0, \beta_1)^T$

$Y^*|X^* \sim N(0, x^{*T}x^*\tau^2 + 1)$

$\underset{n\times 1}{Y} = \underset{n\times p}{X}\underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon}$. $\quad \varepsilon \sim N(0, \sigma^2 I_n)$

σ^2 is known

$\beta \sim N(\beta_0, \Sigma_0)$

ε & β are independent.

Then what is the dist^n of

$\underset{\sim}{Y}|X \simeq MVN(X\beta_0,\ X\Sigma_0 X^T + \sigma^2 I)$

$\underset{\sim}{Y} = \underset{\sim}{\mu} + \underset{\sim}{\varepsilon}$

$\mathrm{corr}(Y_i, Y_j) = \mathrm{corr}(Y_{i'}, Y_{j'})$.

$i \neq j$ $\quad i' \neq j'$

Check $\mathrm{Cov}(Y_i, Y_j | X) \neq \mathrm{Corr}(Y_{i'}, Y_{j'}|X)$ generally.

$i \neq j$ $\quad i' \neq j'$

(specially if $\Sigma_0 = I$)

Check what happens when X is orthonormal.

When the condition holds?

For what value of x is the correlation highest when $\|x\| = 1$.

$e^T \Sigma e$

$A \sim N$

$B \sim N$ $\leftarrow A, B$ independent

then $A + B$

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(σ^2 is known = 1) $\quad Y = X\beta + \varepsilon. \quad \varepsilon \sim N(0, \sigma^2 I_n).$

$$L(\beta; y) \propto \exp\left\{-\frac{1}{2}(Y - X\beta)^T (Y - X\beta)\right\}$$

$$= \exp\left\{-\frac{1}{2}\left(Y^T Y + \beta^T X^T X \beta - 2\beta^T X^T Y\right)\right\}$$

$$\propto \exp\left\{-\frac{1}{2}\left(\beta^T X^T X \beta - 2\beta^T X^T Y\right)\right\}$$

$\mu^2 - 2\mu a \qquad (\mu - a)^2$

$$\exp\left\{-\frac{1}{2}\frac{\sum (y_i - \mu)^2}{\sigma^2}\right\}$$

$$\exp\left\{-\frac{1}{2}\sum (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2\right\}$$

$$\propto \exp\left\{-\frac{1}{2} n(\bar{y} - \mu)^2\right\}$$

$$\exp\left\{-\frac{1}{2}(\beta - \beta_0)^T \Sigma^{-1} (\beta - \beta_0)\right\} = \exp\left\{-\frac{1}{2}\beta^T \Sigma^{-1} \beta^T - 2\beta_0^T \Sigma^{-1} \beta\right\}$$

$$\exp\left\{-\frac{1}{2}\beta^T A \beta - 2\beta^T c\right\}$$

$$= \exp\left\{-\frac{1}{2}(\beta - \alpha_0)^T W (\beta - \alpha_0)\right\}$$

Suppose M is symmetric & invertible.

$x^T M x - 2 b^T x$

$ax^2 - 2bx$

$\left(x - \frac{b}{a^2}\right)^2$

$$(x - c)^T M (x - c) = x^T M x - 2 c^T M x + c^T M c.$$

$$\Rightarrow b^T = c^T M.$$

$$\Rightarrow Mc = b. \Rightarrow \boxed{c = M^{-1} b.}$$

$$c^T M c = b^T M^{-1} b.$$

$$(x - M^{-1} b)^T M (x - M^{-1} b) - b^T M^{-1} b.$$

$\Rightarrow \boxed{x^T M x - 2 b^T x = (x - M b)^T M (\quad)}$.

$\beta \sim N(0, \tau^2 I)$ //

Then, $\boxed{\beta | Y \sim N\left((X^T X + \tau^{-2} I)^{-1} (X^T Y), (X^T X + \tau^{-2} I)^{-1} \right)}$

$\alpha^T (X^T X + \tau^{-2} I) \alpha$.

$\alpha^T (X^T X) \alpha \geq 0$ //

$\alpha^T (\tau^{-2} I) \alpha > 0$ //

$\exp\left\{ -\frac{1}{2} \left(Y^T Y + \beta^T X^T X \beta - 2 \beta^T X^T Y \right) \right\} \exp\left\{ -\frac{1}{2} \frac{\beta^T \beta}{\tau^2} \right\}$

$\tau \to \infty$ // $\left((X^T X)^{-1} X^T Y, (X^T X)^{-1} \right)$

$\tau \to 0$ $\beta | Y \approx$ Point mass at 0 //

$(X^T X)^{-1} X^T Y$

$X^T X +$

MAP is $\left(X^T X + \tau^{-2} I \right)^{-1} X^T$ =

σ^2 is unknown.

$$L = \frac{1}{(2\pi\sigma^2)^{-n/2}} \exp\left\{ - \frac{(y - X\beta)^T (y - X\beta)}{2\sigma^2} \right\}.$$

$$h(\sigma^2) \propto (\sigma^2)^{-(a_0 + 1)} \exp\left(-\frac{b_0}{\sigma^2} \right) = IG(a_0, b_0).$$

$p(\beta, \sigma^2) = p(\sigma^2)\, p(\beta|\sigma^2)$

$\sigma^2 \sim IG$ & $\beta|\sigma^2 \sim N_p(\beta_0, \sigma^2 V_0)$

Do the calculation for the case with $\beta_0 = 0$, $V_0 = I$

$p(\beta, \sigma^2|y)$ $\qquad$ $p(\sigma^2|y) = \int p(\beta, \sigma^2|y)\, d\beta$

Sample: $p(\sigma^2|y)$, then sample $p(\beta|\sigma^2, y)$ $\qquad$ $\sigma^2|y \sim IG(a_n, b_n)$ where $a_n = a_0 + \frac{n}{2}$

$b_n = b_0 + \frac{1}{2}\left[y^T y - \beta_n^T\right.$

prior: $p(\beta, \sigma^2) = p(\sigma^2)\, p(\beta|\sigma^2)$. where $V_n = (X^T X + I)^{-1}$

& $\beta_n = V_n (X^T Y$

$p(\beta, \sigma^2|y) = p(\sigma^2|y)\, p(\beta|\sigma^2, y)$ ((conjugate!

(IG) $\qquad$ MVN

Gibbs sampler:

$p(\beta, \sigma^2|y)$ $\qquad$ Initialise $\sigma^{2(0)}$ $\qquad$ $p(\beta|\sigma^{2(0)}, y)$



(MVN)

β^0

$p(\sigma^2|\beta,y) \propto p(\sigma^2,\beta|y)$

$p(a|b,c,y) \propto p(a,b,c,y) = g(a)$

$\propto g(a)$

$p(\sigma^2|\beta,y) \propto p(\sigma^2$

$p(y$

$= (\sigma^2)^{-n/2} \exp$

$\frac{1}{2}(y-\lambda)) + \frac{1}{2}$

$(\sigma^2)^{-(a_0+1)} \cdot (\sigma^2$

$= (\sigma^2)^{-n/2 + a_0 + 1 + b}$

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$$
\begin{cases}
y = X\beta + \varepsilon \\
\beta \sim N_p(\beta_0, V_0). \\
\varepsilon \sim N(0, \sigma^2 I_n). \\
\sigma^2 \sim IG(a_0, b_0)
\end{cases}
$$

$$
\begin{cases}
Y = X\beta + \varepsilon. \\
\varepsilon \sim N(0, \sigma^2 I_n) \\
p(\beta, \sigma^2) \propto \frac{1}{\sigma^2}
\end{cases}
\Leftarrow
\begin{array}{cc}
y_1 & X_1 \\
\vdots & \vdots \\
y_n & X_n
\end{array}
$$

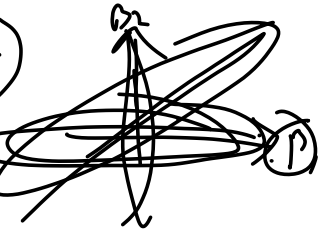
Likelihood: $L(\underline{\beta}, \underline{\sigma^2} | \underline{y}) = \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp\left\{-\frac{(y - X\beta)^T (y - X\beta)}{2\sigma^2}\right\}$

Posterior $\pi(\beta, \sigma^2 | \text{Data}) \propto$ $(\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\}$

Check (with improper prior)

The posterior is proper iff $\int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\} d\beta\, d\sigma^2 < \infty$.

$(X^TX + \lambda I)^{-1} X^T y$



$y = X\beta + \varepsilon$

$\hat{\beta} = (X'X)^{-1} X'Y.$

$\widehat{Var}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$

$\pi(\beta, \sigma^2) \propto \frac{1}{\sigma^2}$ $\pi(\beta_1) \propto 1$

$\pi(\beta,\sigma^2) = \pi(\sigma^2)\pi(\beta|\sigma^2)$ Prior $\beta_1 \sim N(0,\tau^2)$

$\pi(\sigma^2) = \frac{1}{\sigma^2}$ & $\pi(\beta|\sigma^2) = k$

$$Z = \int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{(y-X\beta)^T(y-X\beta)}{2\sigma^2}\right\} d\beta\, d\sigma^2 < \infty ?$$

When $\text{rank}(X) < p$, then is $z = \infty$? X^TX is not invertible.

" " $= p$ ——— $z < \infty$ if $\text{rank}(X) < p$

$$\int_0^\infty (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{-\frac{y^Ty}{\sigma^2}\right\} \int \exp\left\{-\frac{(\beta^TX^TX\beta - 2\beta^TX^Ty)}{2\sigma^2}\right\} d\beta\, d\sigma^2.$$

If $\text{rank}(X) < p$.

Then, $\exists$ a non zero vector v such that $Xv = 0$



Let $u_1 = \frac{v}{\|v\|}$

Complete $(u_1, u_2 .. u_p)$ to be an orthonormal basis & $U = (u_1 \ldots u_p)$

Since U is orthogonal, $U^TU = I_p$.

$\forall \beta \in \mathbb{R}^p$ $\beta = Ut$ where $t \in \mathbb{R}^p$.

$$z = \int_0^\infty \int_{\mathbb{R}^p} (\sigma^2)^{(-\frac{n}{2}+1)} \exp\left\{-\frac{S(Ut)}{2\sigma^2}\right\} dt\, d\sigma^2$$

$X\beta = X(\beta + wv) \forall w$

$\Rightarrow \forall \|y - X\beta\| = \|y - X(\beta + wv)\|$

Define $S(\beta) = \|y - X\beta\|^2$.

$= (\infty (\int^\infty (\sigma^2)^{-(\frac{n}{2}+1)} \cdot (-S^*(t_2 - t_p) \ldots$

$$\int_0^\infty \int_{\mathbb{R}^{p-1}} \int_{-\infty}^{\infty} (\sigma^2)^{-\left(\frac{n}{2}+1\right)} \exp\left\{-\frac{S(t)}{2\sigma^2}\right\} dt_1 \, dt_2 \cdots dt_p \, d\sigma^2 = S^*(t_2, \ldots, t_p)$$

$$= \int_0^\infty \int_{\mathbb{R}^{p-1}} (\sigma^2)^{-\left(\frac{n}{2}+1\right)} \exp\left\{-\frac{S^*(t_2,\ldots,t_p)}{2\sigma^2}\right\} dt_1 = \infty.$$

$\therefore$ rank$(X) < p \Rightarrow$ No proper posterior

When rank$(X) = p$, then is $\pi < \infty$?

$$\int_{\mathbb{R}^p} \exp\left\{-\frac{S(\beta)}{2\sigma^2}\right\} d\beta. \qquad S(\beta) = (y - X\hat{\beta})^T (y - X\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}).$$

where $\hat{\beta} = (X^TX)^{-1}X^Ty$.

$$= \exp\left\{-\frac{SSE}{2\sigma^2}\right\} |X^TX|^{-1/2} (2\pi\sigma^2)^{p/2}.$$

Let $SSE = (y - X\hat{\beta})^T (y - X\hat{\beta})$

$$\int_0^\infty (\sigma^2)^{-\left(\frac{n}{2}+1\right)} (\sigma^2)^{p/2} \exp\left\{-\frac{SSE}{2\sigma^2}\right\} d\sigma^2.$$

$$= \int_0^\infty \sigma^{2-\left(\frac{n-p}{2}+1\right)} \exp\left\{-\frac{SSE}{2\sigma^2}\right\} d\sigma^2.$$

This is the pdf of Inverse Gamma upto a constant.

Now IG pdf doesn't integrate to infinity

$$(IG(a_0, b_0)) \propto (\sigma^2)^{-(a_0+1)} \exp\left(-\frac{b_0}{\sigma^2}\right) \quad a_0 > 0, \; b_0 > 0.$$

$\Rightarrow$ $n > p$ & $SSE > 0$, rank$(X) = p$ are needed for proper posterior

$$\left.\begin{aligned} \beta | \sigma^2, y &\sim N_p(\hat{\beta}, \sigma^2 (X'X)^{-1}) \\ \sigma^2 | y &\sim IG\left(\frac{n-p}{2}, \frac{SSE}{2}\right) \end{aligned}\right\} \rightarrow \text{Matches classical regression results with this prior.}$$

Don't calculate $\beta | y \sim t_{n-p}\left[\hat{\beta}, \frac{SSE}{n-p}(X^TX)^{-1}\right]$

Prior as regularisers.

$y = X\beta + \varepsilon$.
$\varepsilon \sim N(0, \sigma^2 I_n)$.
σ^2 is fixed.
Proper prior.

$p(\beta|y) \propto p(y|\beta)\pi(\beta)$.

$\hat{\beta}_{MAP} = \arg\max_{\beta} p(\beta|y)$.
$= \arg\max_{\beta} \log(p(\beta|y))$
$= \arg\max_{\beta} \{\log p(y|\beta) + \log(\pi(\beta))\}$
$= \arg\min_{\beta} \{-\log p(y|\beta) - \log(\pi(\beta))\}$

For the linear regression likelihood,

$-\log(p(y|\beta)) = \frac{1}{2\sigma^2}\|y - X\beta\|^2 + C$

$\therefore \hat{\beta}_{MAP} = \arg\min_{\beta} \{\frac{1}{2\sigma^2}\|y - X\beta\|^2 - \log \pi(\beta)\}$

Penalty in regularised regression is the negative log-prior.

$\beta_j \sim N(0, \tau^2)$.

$\text{Min}\{\|y - X\beta\|^2 + \lambda\|\beta\|^2\}$.

$\lambda\beta_1^2 + \lambda_2\beta_2^2 + \lambda_3\beta_3^2$
$\lambda\beta_1^2 + \lambda\beta_2^2 + \lambda\beta_3^2$.

$\text{Min}\{\|y - X\beta\|^2 + \lambda\|\beta\|_1\}$.

standardisation is needed.

$\frac{x_1}{s.d(x_1)}$ $\frac{x_2}{s.d(x_2)}$

A1	Kg	Km.
A2	g	Km

First standardise, then.
have $\beta_1, \ldots \beta_p \sim N(0, \tau^2)$
Then Find the equivalence of ridge with β_{MAP}.
when $\beta_1 \ldots \beta_p \sim N(0, \tau^2)$

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MAP estimator for linear regression.

RIDGE

Match

LASSO

~~Fixed and~~ known σ^2.

Ridge: $\arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right\}$ — (1)

$y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$. $\beta_j \overset{iid}{\sim} N(0, \tau^2)$ $j = 1 \dots b$

$\hat{\beta}_{MAP} = \arg\min_{\beta} \left\{ \frac{\|y - X\beta\|^2}{\sigma^2} + \frac{1}{\tau^2} \|\beta\|^2 \right\}$

$= \arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \frac{\sigma^2}{\tau^2} \|\beta\|^2 \right\}$ — (11)

What prior (or what τ) will you choose to get a given value of λ for Ridge regression.

$\tau^2 = \frac{\sigma^2}{\lambda}$ or $\tau = \frac{\sigma}{\sqrt{\lambda}}$

LASSO (known σ): $\arg\min_{\beta} \left\{ \|y - X\beta\|^2 + \lambda_L \|\beta\| \right\}$ — (3)

What will be the Bayesian model (prior of β) such that

$\pi(\beta)$: prior of β $\hat{\beta}_{MAP} = \arg\min_{\beta} \left\{ \frac{\|y - X\beta\|^2}{\sigma^2} - \log \pi(\beta) \right\}$ (equivalent to (3))

$\sigma^2 \cdot \log \pi(\beta) = -\lambda_L \|\beta\|$.

$\Rightarrow \log \pi(\beta) = -\frac{\lambda_L}{\sigma^2} \|\beta\|$. $\Rightarrow \pi(\beta) \propto \exp\left(-\frac{\lambda_L}{\sigma^2} \|\beta\|_1\right)$

$\beta_j \overset{iid}{\sim} \text{Laplace}\left(\frac{\sigma^2}{\lambda_L}\right)$ $j = 1, 2, \dots b$.

$\|\beta\|_1 = \sum_{j=1}^{b} |\beta_j|$

σ^2 unknown: Ridge: $\arg\min \left\{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right\}$.

$y = X\beta + \varepsilon$

$\varepsilon \sim N(0, \sigma^2)$

$\beta \sim N(0, \tau^2)$

$\pi(\sigma^2) \propto \frac{1}{\sigma^2}$

$(\sigma^2, \beta)_{MAP} = \arg\max_{\sigma, \beta} \left\{ \pi(\sigma^2) \pi(\beta) \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ -\frac{\|y - X\beta\|^2}{2\sigma^2} \right\} \right\}$

$= \arg\min_{\sigma, \beta} \left\{ -\log(\pi(\sigma^2)) + n \log(\sigma\sqrt{2\pi}) - \frac{\|y - X\beta\|^2}{2\sigma^2} \right.$

$\|\beta\|^2$)

$\beta|\sigma \sim N(0, \frac{\sigma^2}{\tau^2})$

$\times \pi(\sigma^2)$

$= \arg\min_{\sigma,\beta} \left\{ C(\sigma) - \frac{\|y-X\beta\|^2}{2\sigma^2} - \frac{\|\beta\|^2}{2\tau^2} \right\}$

MAP for this model gives you the exact answer for β as Ridge Regression

$y = X\beta + \varepsilon$,

$\varepsilon \sim N(0, \sigma^2 I_n)$

$\sigma^2 \sim \pi(\sigma^2)$

$\beta|\sigma^2 \sim N(0, \frac{\sigma^2}{\tau^2} I)$

$\arg\min_{\sigma,\beta}$ (For every value of σ the loss fn. w.r.t β minimizes.)

$\min \{ \|y-X\beta\|^2 + \lambda \|\beta\| \}$ $\|y-X\beta\|^2 + \lambda\|\beta\|^2$ $\|y-X\beta\|^2 + \lambda\|\beta\|$

$C(\sigma) e^{-\frac{Q(\beta)}{\sigma}}$

$y = X\beta + \varepsilon$

$\varepsilon \sim N(0, \sigma^2 I_n)$

$\sigma^2 \sim \pi(\sigma^2)$

$\beta|\sigma^2 \sim \text{Laplace}(0, \frac{\sigma^2}{\lambda})$

Elastic net

$p(\beta|\sigma^2) \propto \exp\left\{ -\frac{(\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2)}{2\sigma^2} \right\}$ $\hat{\beta} = \arg\min \{ \|y-X\beta\|^2 + \lambda\|\beta\|_1 + \lambda_2\|\beta\|^2 \}$

Bayesian Lasso

$y|\beta,\sigma^2 \sim N_n(X\beta, \sigma^2 I_n)$

$\beta_j|\sigma \sim \text{Laplace}(0, \frac{\sigma}{\lambda})$

$p(\beta_j|\sigma,\lambda) \sim \frac{\lambda}{2\sigma} \exp\left(-\frac{\lambda|\beta_j|}{\sigma}\right)$

Use this property to find Gibbs Sampler

If $\tau_j^2|\lambda \sim \text{Exp}(\frac{\lambda^2}{2})$

& $\beta_j|\tau_j^2,\sigma^2 \sim N(0, \sigma^2\tau_j^2)$

then, $\beta_j|\sigma,\lambda \sim \text{Laplace}(0, \frac{\sigma}{\lambda})$

(M) Normal & scaled-inverse-χ^2

$\sigma \sim (ik)$

unconditional it was.

$\mu \sim t$

$y|\beta,\sigma^2 \sim N(X\beta, \sigma^2)$

$\beta_j|\tau_j^2,\sigma^2 \overset{ind}{\sim} N(0, \sigma^2\tau_j^2)$

$\tau_j^2 \overset{iid}{\sim} \text{Exp}(\lambda^2/2)$

$\sigma^2 \sim \text{IG}(a_0, b_0)$

$\beta|\sigma^2, y, \tau^2 \sim$ [ ]

$\sigma^2|\beta, y, \tau^2 \sim$ [ ]

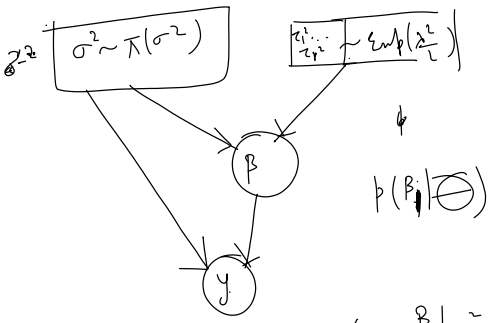
$\tau^2|\sigma^2, \beta, y \sim$ [ ]

Joint posterior: $p(\sigma^2, \tau^2, \beta|y) \propto p(\sigma^2, \tau^2, \beta, y)$

$= p(\sigma^2)\, p(\tau^2)\, p(\beta|\tau^2, \sigma^2)\, p(y|\beta, \sigma^2)$

$= p(\sigma^2) \left(\prod_{j=1}^{p} p(\tau_j^2)\right) \left(\prod_{j=1}^{p} p(\beta_j|\tau_j^2, \sigma^2)\right) \prod_{i=1}^{n} p(y_i|\beta, \sigma^2)$

$\rightarrow (*)$



$p(\beta_j|\ominus) \propto p(*)$

$\propto p(\beta_1|\tau_1^2, \sigma^2)\, p(y|\beta, \sigma^2)$

$\beta|\sigma^2 \sim N_p(V_\beta X^T y, \sigma^2 V_\beta)$ where $V_\beta = (X^TX + D_\tau^{-1})^{-1}$

where $D_\tau = \begin{pmatrix} \tau_1^2 & 0 & & \\ 0 & \tau_2^2 & 0 & 0 \\ & 0 & \ddots & \\ & & & \tau_p^2 \end{pmatrix}$

$\sigma^2|\beta, \tau^2, y \sim \text{IG}(a_n, b_n)$

$\tau_j^2|\sigma^2, \beta, y \sim$ [ ] (Metropolis-Hastings in this step)

MH within Gibbs

No Class this week

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E

LASSO — RIDGE

Regularisation via Bayesian inference.

LASSO/RIDGE are MAP solution under appropriate priors

Penalty

Can be used to provide information which may help us in other way (such as regularisation.

Variable selection in linear regression.

→ Many covariates are there but someof them have no effect ($\beta_j = 0$ for some j. such covariates can be more than one).

Classical setup: Fit model.

Based on p-value, you do variable selection.

(10) → (1) Forward selection | Backward selection → How to do in Bayesian setup.

$y = X\beta + \varepsilon$ which includes all covariates.

Then remove each of the covariate and check which has minm AIC

You compare full model and reduced model based on

M1 vs M2 : BF

$y = X\beta + \varepsilon$

Spike & Slab prior

$y \mid X, \beta, \sigma^2 \sim N(X\beta, \sigma^2 I_n)$

$\sigma^2 \sim IG(a_0, b_0)$

$\beta_j \mid \gamma_j = \begin{cases} 0 & \text{if } \gamma_j = \\ N(0, \sigma^2 \tau_1^2) & \text{if } \gamma_j \end{cases}$

$\gamma_j \sim Ber(\pi)$ $\forall j$ independently

$P(\gamma_j = 0 \mid X, Y)$

$\beta_j \mid \gamma_j, \sigma^2 \sim \gamma_j \delta_0 + (1-\gamma_j) N(0, \sigma^2 \tau_1^2)$ $\quad \beta_0 \sim N(0, \sigma^2 \tau_1^2)$

Gibbs sampling to estimate the parameters?

$\beta_0, \beta_1, \ldots \beta_p, \gamma_1, \ldots \gamma_p, \sigma^2$

Spike and slab prior for $\beta_1, \ldots \beta_p$, not for intercept.

$\beta_0^{(t+1)} \mid \beta_1^{(t)}, \beta_2^{(t)}, \ldots \gamma_1^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$\beta_1^{(t+1)} \mid \beta_0^{(t+1)}, \beta_2^{(t)}, \ldots \gamma_1^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$\gamma_1^{(t+1)} \mid \beta_0^{(t+1)}, \ldots \beta_p^{(t+1)}, \gamma_2^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$(\beta_1, \beta_2, \gamma_1, \gamma_2, \sigma^2)$

$R^{(t+1)} \mid \beta_1^{(t)}, \gamma_1^{(t)} = 1, \gamma_2^{(t)}, \sigma^{2(t)}$

$\sigma^{2(t+1)} \mid \beta_2^{(t)}, \ldots, \beta_p^{(t)}, \gamma_1 \ldots \gamma_p^{(t)}$ |

$\Rightarrow \beta_1^{(t+1)} = 0$ w.p. 1.

$\gamma_1^{(t+1)} \mid \beta_2^{(t)}, \beta_1^{(t+1)}=0, \gamma_2^{(t)}, \sigma^{2(t)}$

$\gamma_1^{(t+1)} = 1$ w.p. 1

This violates irreducibility of the Markov chain & hence the sampler will not converge to the ~~the~~ target distn.

$\beta_1^{(t+1)} \mid \beta_2^{(t)}, \gamma_1^{(t)}=0, \gamma_2^{(t)}, \sigma^{2(t)}$.

Then $\beta_1^{(t+1)} \neq 0$ w.p. 1. $\quad \gamma_1^{(t+1)} \mid \beta_1^{(t+1)} \neq 0, \gamma_2^{(t)}, \sigma^{2(t)}, \beta_2 = 0$ w.p. 1

$\beta_j \mid \gamma_j \sim \gamma_j N(0, \tau_0^2) + (1-\gamma_j) N(0, \sigma^2 \tau_1^2)$ where τ_0 is very small. (Jiehering continuous spike & slab)

Gibbs sample (A, B, C) $\quad A \mid B, C$; $B \mid A, C$; $C \mid A, B$.

A, B, C need not be ~~scalar~~

$(\beta_1, \beta_2, \gamma_1, \gamma_2, \sigma^2)$

$\leftarrow (\beta_1, \gamma_1) \mid \beta_2, \gamma_2, \sigma^2)$

$(\beta_2, \gamma_2) \mid \beta_1, \gamma_1, \sigma^2)$

$\sigma^2 \mid \beta_1, \gamma_1, \gamma_2, \beta_2$

~~$(\gamma_1 \mid \beta_2, \gamma_2, \sigma^2)$~~ then $(\beta_1 \mid \gamma_1, \beta_2, \gamma_2, \sigma^2)$

Collapse Gibbs sampler $\Rightarrow$

$(\gamma_j, \beta_j) \mid \gamma_{-j}, \beta_{-j}, \sigma^2 \quad (\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$ & $(\beta_j \mid \gamma_j, \gamma_{-j}, \beta_{-j}, \sigma^2)$

$\gamma_j \in \{0, 1\}$ $\quad P(\gamma_j = 0 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$ & $P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$\hat{p}_j$

$u \in (0, 1)$

$u \sim \text{Uniform}(0,1)$ if $u < \hat{p}_j$ then $\gamma_j = 0$

else $\gamma_j = 1$.

When $\gamma_j = 1 \Rightarrow \beta_j = 0$.

When $\gamma_j = 0$ then we generate β_j from $(\beta_j \mid \gamma_j, \beta_{-j}, \sigma^2)$

$P(\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$= \int P(\gamma_j, \beta_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \, d\beta_j$ ~~$= P(\gamma_j, \beta_j \mid \gamma_{-j}, \beta_{-j})$~~

$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$P(\gamma, \beta \mid \gamma, \beta_{-j}) \propto P(\gamma_j) P(\beta_j \mid \gamma_j) P(y \mid \beta, \sigma^2)$

$$= P(\gamma_j)\, P(\beta_j \mid \gamma_j)\, \frac{1}{(\sigma\sqrt{2\pi})^n} \exp - \left\{ \sum_{i=1}^{n} \frac{\left(y_i - \underbrace{\sum_{k \neq j} x_{ik}\beta_k}_{z_i} - x_{ij}\beta_j \right)^2}{2\sigma^2} \right\}$$

$$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \pi \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ - \frac{\sum z_i^2}{2\sigma^2} \right\}$$

Prior for γ_j is $\bar{\pi}$

$$P(\gamma_j = 0 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \int (1-\bar{\pi}) \frac{1}{\sigma\tau_1\sqrt{2\pi}} \exp - \left\{ \frac{\beta_j^2}{2\sigma^2\tau_1^2} \right\} \frac{1}{(\sigma\sqrt{2\pi})^n} \exp - \left\{ \frac{\sum (z_i - x_{ij}\beta_j)^2}{2\sigma^2} \right\} d\beta_j$$

$$= (1-\bar{\pi})$$

$z_j = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$

$z_j \mid \beta_j \sim N(x_j \beta_j, \sigma^2 I_n)$ — ①

$\beta_j \sim N(0, \sigma^2\tau_1^2) \Rightarrow x_j\beta_j \sim N(0, \sigma^2\tau_1^2 x_j x_j^T)$ — ②

$N(0, \sigma^2(I_n + \tau_1^2 x_j x_j^T))$

$= (1-\bar{\pi})\, f(z_i; \sigma^2(I_n + \tau_1^2 x_j x_j^T)) = a_j$

$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \frac{\bar{\pi}}{(\sigma\sqrt{2\pi})^n} \exp\left\{ - \frac{z_i^2}{2\sigma^2} \right\} = b_j$

then, $P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) = \frac{b_j}{a_j + b_j} = p_j$

$\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2 \sim \text{Ber}(p_j)$

if $u < p_j$, then $\beta_j = 0$

else $\left(\beta_j \mid z_j, \sigma^2, \gamma_j = 0 \right) \sim N(m_j, v_j)$ (γ|w)

where $m_j = \frac{\tau_j^2 x_j^T z_j}{1 + \tau^2 x_j^T x_j}$

$v_j = \frac{\sigma^2\tau^2}{1 + \tau^2 x_j^T x_j}$

$(\gamma_1, \beta_1 \mid \gamma_{-1}, \beta_{-1}, \sigma^2)$

$(\gamma_2, \beta_2 \mid \gamma_{-2}, \beta_{-2}, \sigma^2)$

$\vdots$

→ Updated

$(\sigma^2 \mid \gamma, \beta)$

$\beta_j \mid \gamma_j \underset{\sigma^2}{\sim} \gamma_j S_0 + (1-\gamma_j) N(0, \sigma^2\tau_1^2)$

$\pi(\sigma^2) \propto \frac{1}{\sigma^2}$ $\pi(\sigma^2 | \underline{\gamma}, \underline{\beta}) \propto \pi(\sigma^2) \prod_{j=1}^{p} \pi(\gamma_j) \pi(\beta_j | \gamma_j, \sigma^2) \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ \frac{\|y - X\beta\|^2}{2\sigma^2} \right\}$

$\beta_j | \sigma^2 \sim N(0, \sigma^2 c^2)$

$\Downarrow$

$\pi(\sigma^2 | \beta) = IG\left(\frac{n+p}{2}, \frac{\|y - X\beta\|^2 + \sum \beta_j^2 / c^2}{2}\right)$

$\sigma^2 | \beta, \gamma, y \sim IG\left(\frac{n+k}{2}, \frac{\|y - X\beta\|^2 + \sum \beta_j^2 / c^2}{2}\right)$

where k = no. of γ's which are 0.

$P(\beta_j \neq 0 | \text{Data})$

$1 - \frac{\sum \gamma_j}{2000}$

10000

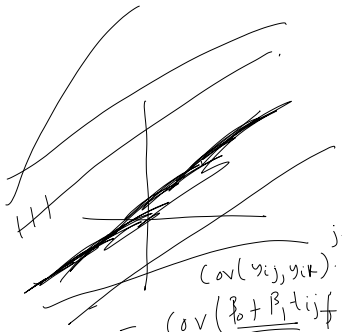
2000 ✓

$(\beta_j, \gamma_j, \sigma^2)$ - - - ($\forall j = 1(1)p$

Random-effects

$i = 1, 2, \ldots n$ (patients)

For each i, we have n_i observations



$= cov(\beta_0 + \beta_1 t_{ij} + b_i + \varepsilon_{ij}, \beta_0 + \beta_1 t_{ik} + b_i + \varepsilon_{ik})$

$= \tau^2$

$cor(y_{ij}, y_{ik}) = \frac{\tau^2}{\tau^2 + \sigma^2}$ & $cov(y_{ij}, y_{mn}) = 0$ if $i \neq m$

$y_{ij} = \underbrace{\beta_0 + \beta_1 t_{ij}}_{\text{Fixed effect}} + b_i + \varepsilon_{ij}$

$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$

$b_i \overset{ind}{\sim} N(0, \tau^2)$

β_0: Population intercept

β_1: Population slope

b_i: random intercept for ith individual

σ^2: the within individual error variance

τ^2: Between-individual error variance

$(\beta_0, \beta_1, \sigma^2, \tau^2)$

Prior over the parameters

Δ then you try to find posterior

$cov(y_i, y_j) = 0$

$cov(y_{ij}, y_{ik}) = \tau^2$ $j \neq k$

$V(y_{ij})$ --